MILITARY SPECIFICATION

INDICATOR, ATTITUDE-DIRECTOR, REMOTE, ID 1329A(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers design and all performance requirements for two types of a remote attitude-director indicator, which is a component of the integrated flight instrumentation system. The indicator displays pitch and roll attitude and azimuth information with full freedom about each axis and flight director information controlled by remote circuits.

1.2 Classification. The indicators are the following two types:

1.2.1 ID 1329A - AC all axises drive motor
1.2.2 ID 1329A/A - DC all axises drive motor

1.3 Associated equipment. The remote attitude-director indicator manufactured in accordance with this specification is a component of an integrated flight instrumentation system conforming to MIL-S-23535. The other components are the horizontal situation indicator manufactured in accordance with MIL-I-23366 and the Flight Director Computer manufactured in accordance with MIL-C-23367.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC 6610

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SPECIFICATIONS

FEDERAL

BB-N-411	-	Nitrogen, Technical.
DD-G-451	-	Glass, Float or Plate, Sheet, Figured (Flat, for Glazing, Mirrors and Other Uses).
PPP-C-795	-	Cushioning Material, Flexibale, Cellular, Plastic Film for Packaging Applicantions.
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-C-1797	-	Cushioning Material, Resilient, Low Density Unicellular, Polypropylene Foam
PPP-C-1842	-	Cushioning Material, Plastic, Open Cell for Packaging Applications).
QQ-C-320	-	Chromium Plating (Electrodeposited).
QQ-N-290	-	Nickel Plating (Electrodeposited).
QQ-P-416	-	Plating, Cadmium (Electrodeposited).

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MIL-P-116	-	Preservation, Methods of.
MIL-B-131	-	Barrier Materials, Watervaporproof, Greaseproof, Flexible Heat Sealable.
MIL-C-675	-	Coating Of Glass Optical Elements (Antireflection).
MIL-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on.
MIL-E-5400	-	Electronic Equipment, Aerospace, General Specification for.
MIL-T-5422	-	Testing, Environmental, Airborne Electronic Equipment.
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-L-6085	-	Lubricating Oil, Instrument, Aircraft, Low Volatility.
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for.
MIL-M-7793	-	Meter, Time Totalizing.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-T-18303	-	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-I-23366	-	Indicator, Horizontal Situation, ID-1013/A
MIL-C-23367	-	Computer, Flight Director, Aircraft.
MIL-S-23535	-	System, Integrated Flight Instrumentation.
MIL-L-25467	-	Lighting, Integral, Red, Aircraft Instrument, General Specification For.
MIL-L-26482	-	Connectors, Electrical (Circular, Miniature Quick Disconnect, Environment Resisting) Receptacles and Plugs, General Specification for.

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MIL-G-81559 - Gyroscope Assemblies and Attitude and Directional Reference Instructions for Aircraft; Packaging of

STANDARDS

FEDERAL

FED-STD-595 - Colors.

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MIL-STD-129 - Marking For Shipment And Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-704 - Electric Power, Characterisitics.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIl-STD-794 - Part and Equipment, Procedures for Packaging and Packaging of
MIL-STD-889 - Dissimilar Metals.
DOD-STD-1866 - Soldering Process, General (Nonelectrical).
MS18028 - Indicator, Attitude - Director, Remote.
MS33558 - Numerals and Letters, Aircraft Instrument.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging, Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generallly available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Parts.

3.2.1 Commercial parts. Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts and cotter pins, having suitable properties, shall be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts. With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical materials. Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic materials. Non-magnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Protective treatment. When materials are used in the construction of the indicator which is subjected to corrosion in environmental conditions likely to occur during normal service life, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of environmental conditions, shall not be used.

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3.3.3.1. Metals. Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt environment or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.2 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected aganst electrolytic corrosion by means of protective coatings or hermetic sealing.

3.3.3.3 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the acquiring activity shall be used. (see 6.2.1)

3.3.3.4 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625. Dial plates, small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.5 Iron and steel parts. Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium or nickel plated in accordance with QQ-C-320 or QQ-N-290, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

3.3.3.6 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the acquiring activity (see 6.2.1). If used in a hermetically sealed case, fungicidal treatment will not be necessary.

3.3.3.7 Fumes and vapors. Materials used in the construction of the indicator shall not produce corrosive, deleterious or toxic fumes or vapors under the conditions specified herein.

3.3.4 Lubrication. The lubricating oil used in the indicator shall conform to MIL-L-6085. The amount of lubricating oil used shall be sufficient for at least 1000 hours of operation.

3.4 Design and construction. The indicator shall be designed and constructed so that the complete mechanism consisting of all components, except the pitch trim knob assembly and lighting components, as required, shall be removable from the case as a unit. It shall be so constructed that no parts shall work loose in service and shall withstand normal shocks, vibrations and such other conditions as are incident to shipping, storage, installation and service without failure. The indicator shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

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3.4.1 Maintainability. The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance and reassembly using those tools and items of maintenance which are normally available as commercial standard equipment without requiring special tools and fixture.

3.4.2 Case. The case shall conform to MS18028, shall be made of non-ferrous, low density metal, uniform in texture, with a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service. The pitch trim knob shall be 3/4 inch in diameter and one inch long. The case shall be isolated from all electrical circuits within it.

3.4.2.1 Hermetic sealing. The case shall provide a hermetically sealed enclosure for the interior mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the acquiring activity (see 6.2.1). The sealing of the case shall not be dependent upon any material which shall be adversely affected by any atmosphere to which the indicator may be subject in normal use encountered in military aircraft.

3.4 2.2 Filling medium. The filling medium shall be a mixture of 95 percent helium and 5 percent carbon dioxide or a mixture of gases approved by the acquiring activity (see 6.2.1). The nitrogen used shall conform to BB-N-411, Type I, Class 1, Grade A. The filling medium shall be of at least 98 percent purity, free of dust particles and shall not contain more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The case shall be filled to an absolute pressure of 1 ± 0.1 atmosphere.

3.4.2.3 Fogging. There shall be no fogging of the cover glass when the indicator is energized for a minimum of 1 hour in a 71°C ambient temperature and the indicator face then rubbed with an ice cube for a 1 to 2 minutes. (see 4.6.32 1)

3.4.3 Component parts. The indicator shall consist of the parts and mechanisms necessary to provide and operate a roll, pitch and azimuth display; a displacement indicator; a rate of turn indicator; a slip indicator and appropriate warning flags, as specified herein and illustrated in Figures 1, 2, 3 and 4.

3.4.3.1 Time totalizing meter. The indicator shall be designed to include, within its case, a time totalizing meter conforming to MIL-M-7793.

3.4.3.2 Attitude display. The attitude display shall basically consist of a bank angle dial, bank indices, a miniature airplane, and a sphere background which is supported and rotated about three servo driven gimbal axes. The overall arrangement of the attitude display shall be in accordance with Figure 1.

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3.4.3.3 Dial masks. A mask shall be attached to each side of the bank scale dial in accordance with Figures 1, 2, 3 and 4. The mask on the left shall also be the scale for the displacement pointer, (see 3.4.7.4). The surface of the scale shall follow the arc of the displacement pointer movement. The right mask shall be the shield behind which the vertical pointer is driven, (see 3.4.7.1), the mask shall be designed so that the displacement pointer can be driven out of view to the top as shown in Figure 1.

3.4.3.4 Mechanism mask. A mask which is attached to the roll gimbal and extends essentially from the bank dial to the sphere shall be incorported. The visibible surface of this mask shall be shaped as the surface of the frustrum of a cone. The large diameter shall be forward. The edge of the mask shall be hidden by the inner diameter of the bank scale.

3.4.3.5 Bank indices. Bank indices shall be fixed to and rotate with the roll gimbal. They shall be in accordance with Figures 1 and 2. At zero roll indication, one shall be adjacent to and aligned with the zero roll marking on the bank angle dial. The other index shall be at the 180 degree roll position.

3.4.3.6 Miniature airplane. The miniature airplane shall be in accordance with Figure 2. The design of the indicator shall be such that the miniature airplane shall be secured to the bank scale at the zero pitch position.

3.4.4 Gimbal system. The gimbal system shall consist of the following:

3.4.4.1 Roll gimbal. The roll gimbal shall be the outer gimbal and shall be capable of rotating 360 degrees. The axis of the gimbal shall be perpendicular to the face of the indicator and in line with the center of the sphere. The gimbal shall be driven by the roll servo mechanism. The gimbal shall be designed to mount the bank indices and the pitch gimbal.

3.4.4.2 Roll trim. Means shall be incorporated to adjust the roll gimbal a minimum of ± 3 degrees. The adjustment shall be located on the amplifier case.

3.4.4.3 Pitch gimbal. The pitch gimbal shall be suspended in the roll gimbal and shall be capable of rotating 360 degrees. The axis of the gimbal shall be at right angles to the roll gimbal axis and in line with the roll axis. The gimbal shall be driven by the pitch servo mechanism. The azimuth mechanism shall be mounted on the pitch gimbal.

3.4.4.4 Pitch trim. Means shall be incorporated to adjust the position of the sphere in pitch by means of a pitch trim knob mounted on the lower right hand side of the face of the indicator. The knob shall have approximately one-fourth turn of freedom in the counterclockwise direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob shall be rotated to the stop in the clockwise direction one-half turn, the sphere shall rotate to deflect

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the horizon line upward to indicate between 10 and 20 degrees dive. When the knob is rotated to the stop in the counterclockwise direction, the sphere shall rotate to deflect the horizon line downward to indicate between 5 and 10 degrees climb. The servo system shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set, but it shall move smoothly without evidence of binding or erratic movement. The internal design of the indicator shall be such that any pitch trim input shall be automatically faded out as the sphere shall be driven to the extreme pitch indications and the sensing of the trim signal reversed so as to correspond to the side of the sphere showing when the sphere shall be driven through the 90 degree pitch positions. The fade out shall occur as the sphere is driven from the 20 to 80 degree pitch indications and shall fade in as the indicated pitch angle changes from 80 to 20 degrees. It shall occur on both sides of the sphere for both climb and dive indications.

3.4.4.5 Azimuth mechanism. The azimuth mechanism shall consist of a shaft and the azimuth servo mechanism and shall be capable of rotating 360 degrees. The shaft shall be mounted so that it shall be perpendicular to the pitch gimbal axis and, at the zero pitch attitude, perpendicular to the roll gimbal axis. The shaft shall support the two sphere halves at their centers. The shaft shall be driven by the azimuth servo mechanism.

3.4.4.6 Sphere. The sphere which forms the background of the display shall consist of two half spheres. The half spheres shall be fastened to the end of the azimuth shaft. The sphere shall rotate with the roll gimbal for roll indications, rotate in relation to the roll gimbal about the pitch axis for pitch indication and rotate in relation to the pitch and roll axes about the azimuth axis for azimuth indications. The sphere shall be driven in pitch so that the horizon line goes above the minature airplane for dive maneuvers and below for climb maneuvers and shall be driven in azimuth so that it moves to the left for right turns. The gimbal shall rotate counterclockwise for right banks. All marking and coloring of the sphere presentation shall be in accordance with Figures 5 and 6.

3.4.5 Indicator synchro control transformer. The synchro control transformer on the indicator axes shall be high impedance Eclipse-Pioneer Type AY500-5 synchros or equal The design of the servo circuit shall be such that when one to four indicators are connected to a single test transmitter synchro per axis no noticeable change in performance shall be noted. All control transformer stator leads shall be electricaly isolated from other circuitry in the case and from the case.

3.4.5.1 Indication zeroing and sensing. When the pitch, roll and azimuth servo control transformers of the indicator are connected to standard test transmitters in accordance with Table I with the transmitters for pitch and roll set at the 180 degrees dial reading and the transmitter for azimuth set at the zero degree dial reading, the indications shall come to rest at zero. Increasing dial readings of the roll, pitch, and azimuth test transmitters shall produce right roll, pitch up and right turn indications. Decreasing dial readings shall produce opposite indications.

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3.4.5.2 Attitude actuation. The sphere shall be servo rotated about the pitch, roll, and azimuth axes by means of geared servo motors (until a null is attained between the control transformer synchros on the indicator axes and the transmitter synchros in the pitch, roll, and azimuth displacement reference). The freedom of sphere rotation about any axis shall not be limited.

3.4.5.3 Follow-up rate. The rate at which the motors drive the gimbals shall be not less than 300 degrees per second in azimuth and roll and not less than 90 degrees per second in pitch.

3.4.5.4 Accuracy. The pitch, roll and azimuth indications shall agree with the input signals from their respective test transmitter synchros to within ± 1/2 degree for all positions through 360 degrees, and the indication of each variable shall be independent of the position of the other two.

3.4.5.5 Follow-up operation. The electrical and mechanical design of the indicator shall be such that in the process of follow-up, the indications in pitch, bank, and azimuth shall move smoothly without hunting, oscillation, or jumping greater than 1/4 degree total amplitude in pitch, bank, or azimuth when connected to a standard test transmitter. There shall be no sustained electrical or mechanical oscillations in any axis greater than 1/4 degree total amplitude at any indication. There shall be no overshooting greater than three degrees with a step input equal to the maximum follow-up lag.

3.4.5.6 Amplifiers. Amplifiers shall be provided when the indicator is not a part of the AJB-3A system (bomb release computing set). The amplifiers provided as a part of the indicator shall be transisterized amplifiers. The amplifiers shall be designed with redundant circuits such that the indicator shall continue to operate, though with reduced performance, should partial failure of the amplifier occur. The amplifiers shall be housed in a package separate from the indicator proper such that it can be physically attached to the rear of the indicator proper or mounted remotely. The design of the amplifier shall be such that it shall meet the requirements of this specification when mounted directly to an airframe without shock mounts, as well as, when mounted on the indicator. The amplifier shall be so designed that each channel shall be a separate module within the amplifier case. The amplifier shall be so designed that each channel is a separate module within the amplifier case. All channels shall be interchangeable, if practicable.

3.4.5.7 Adapter. When the indicator is used with the AJB-3A system, the amplifier shall not be required, but shall be replaced by an adapter containing signal mixing circuitry conforming to Figure 7 relaying and associated components required for operation with the AJB-3A system.

3.4.5.8 Zeroing adjustments. Means shall be incorporated to externally

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adjust the position of the sphere ± 3 degrees minimum in pitch and roll. The initial setting of the pitch and roll indication at the time of delivery of the indicator shall indicate zero pitch and roll within ± 1/2 degree when the indicator is connected to a test transmitter set to the 180 degrees dial reading. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The roll adjustment shall be on the amplifier case. It shall be adequately marked and easily settable with the amplifier mounted remotely or on the indicator proper.

3.4.6 Inclinometer. The inclinometer shall be incorporated into the display in accordance with Figures 1, 3 and 4.

3.4.6.1 Tube. The glass of the inclinometer tube shall be made of clear annealed glass tubing free from any flaws that will seriously affect the readability of the inclinometer. The inside of the tube shall be smooth and uniform so that the ball shall roll freely, except for the damping of the liquid. The arc length of the visible portion of the tube shall be approximately 1-1/4 inches. The radius of curvature of the tube shall be approximately 3-1/2 inches. The background of the tube shall be painted with lusterless white material matching Color No. 37875 of FED-STD-595.

3.4.6.2 Ball. The ball shall be a highly polished accurate 7/32 inch black sphere. A minimum of 2/3 of the ball shall remain in view at either extreme of the tube, when viewed from 12 inches in front of the zero mark of the slip indicator at an angle of 30 degrees with the horizontal plane described by the base of the indicator.

3.4.6.3 Damping liquid. The damping liquid used shall be sufficiently colorless to preclude interference with the visibility of the ball under day light and night lighting conditions and under normal and extreme temperatures.

3.4.7 Flight director display. The flight director display shall consist of a vertical pointer, a horizontal pointer, a pointer to show replacement from the commanded flight path, and two signal level flag alarms. These pointers and flags shall be driven by meter movements and shall be in accordance with Figures 1, 2 and 3. The design of the pointer meter movements shall be such that when not energized these pointers are maintained to their adjusted zero positions while under all flight conditions. All meter movement leads shall be electricallly isolated from other circuitry in the case and from the case. Unless otherwise specified, all meter movements shall operate as specified herein when used singularly, when connected in parallel with a like meter movement, or when connected in parallel with a 1,000 ohms load to a signal source impedance between 350 and 1,000 ohms.

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3.4.7.1 Vertical pointer. The vertical pointer shall be mounted on a meter movement which is pivoted at a point approximately 4-1/2 inches behind the pointer in such a manner that, with the indicator in normal operating position, the pointer shall remain vertical at zero center or at any point of deflection to the right or left of center, and shall deflect right or left in agreement with the polarity of direct current received from the flight director computer. An external mechanical zero adjustment shall be provided. The following conditions shall be observed:

a. The flight director computer referred to herein shall be designed so that its current to the pointer movements is limited to 2.5 ma maximum when in a mode where the pointers are being used and it shall provide 10 to 13.5 ma of the proper polarity to either or both pointer movements when these pointers are to disappear.

3.4.7.1.1 Deflection. The design shall be such that the deflection of the pointer to the left of the center position is limited to one inch by a physical stop and the maximum deflection to the right will carry the pointer behind the mask on the right side of the display. The direction of deflection for a given input signal shall be as specified in Table I.

3.4.7.1.2 Response. The vertical pointer response shall be linear in degrees with respect to current within 7.5 percent of the proportionate full scale value. A deflection of 7/8 of an inch, measured from the center of the dial to the pointer along the horizontal line, shall require 2.2 ma ± 7.5 percent, and 2.5 ma shall not deflect the pointer by more than 1-1/16 inches. An application of 10 ma of the proper polarity shall drive the pointer from view behind the mask to the right. The meter movement shall not be damaged when 10 to 13.5 ma is applied as a step input.

3.4.7.1.3 Resistance. The resistance across the input terminals of the vertical pointer shall be 1000 ohms ± 3 percent at 25°C.

3.4.7.1.4 Damping. The vertical pointer mechanism shall be damped so that there is no more than 1-1/2 percent overshoot.

3.4.7.1.5 Response time. The response time of the vertical pointer shall be a maximum of 1/3 second. Response time is defined as the time required for the pointer to reach 90 percent of its final indication, and the time shall be counted from the instant of application of the current. The pointer shall be considered as having come to apparent rest when it has reached within ± 1 percent of the scale length of the actual rest point.

3.4.7.2 Horizontal and vertical pointer flag alarms. The horizontal and vertical pointer flag alarms shall be operated by a suppressed zero type of mechanism which shall hold the flags against stops in a position shown in Figure 1 in the absence of current, or when the current application is below a predetermined value as specified below.

3.4.7.2.1 Deflection. The horizontal and vertical flag alarms shall deflect out of view with the input signal polarity of Table I connected to a voltage which is polarized plus and minus, respectively.

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3.4.7.2.2. Response. The alarm flags shall leave their visible-position stop with a current application of 180 microamperes or more. The flag alarms shall disappear from view with a current application of 245 microamperes. Response of the flag alarm mechanisms shall be linear in degrees with respect to current. The application of 380 microamperes as a step shall not damage the meter movements.

3.4.7.2.3 Resistance. The resistance across the input terminals of the horizontal and vertical pointer flag alarms shall be 1,000 ohms ± 3 percent at 25°C.

3.4.7.3 Horizontal pointer. The horizontal pointer shall be positioned directly behind and as near as practicable to the vertical pointer. The horizontal pointer shall be mounted on a meter movement which is pivoted at a point approximately 4-1/2 inches behind the pointer. With the indicator in normal operating position, the pointer shall move in such a manner that it remains horizontal at zero center or at any point of deflection up or down from center, and shall deflect up or down in agreement with the polarity of direct current received from the flight director computer. An external mechanical zero adjustment shall be provided.

3.4.7.3.1 Deflection. The maximum deflection upward shall be physically limited to 1-1/16 inch. The design shall be such that the pointer can be deflected downward (1-15/16 inches minimum) until it disappears from view behind the mask. The direction of deflection for a given input signal shall be as specified in Table I.

3.4.7.3.2 Response. The horizontal pointer response shall be linear in degrees with respect to current to within 7.5 perent of the proportionate full scale value. A deflection of 7/8 inch measured from the center of the dial to the pointer along a vertical line shall require 2.2 ma ± 7.5 percent and 2.5 ma will not deflect the pointer by more than 1-1/16 inch. An application of 10 ma of the proper polarity shall drive the pointer from view behind its mask at the bottom of the display. The application of 13.5 ma as a step input to drive the pointer from view shall not damage the meter movement.

3.4.7.3.3 Resistance. The resistance across the input terminals of the horizontal pointer shall be 1000 ohms ± 3 percent at 25°C.

3.4.7.3.4 Damping. The horizontal pointer mechanism shall be damped so that there is no more than 1-1/2 percent overshoot.

3.4.7.3.5 Response time. The response time of the horizontal pointer shall be a maximum of 1/3 second. Response time is defined in 3.4.7.1.5.

3.4.7.4 Displacement pointer. The displacement pointer and scale (see 3.4.3.3) shall be in accordance with Figure 3. The pointer shallbe mounted on a meter movement which is pivoted at a point approximately 4-1/2 inches behind the pointer. With the indicator in normal operating position, the pointer shall move in such a manner that it shall remain horizontal at zero center or at any point of deflection up or down from center, and shall

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deflect up or down in agreement with the polarity of direct current received from the flight director computer. An external mechanical zero adjustment shall be provided. The pointer shall run adjacent to the displacement scale as shown in Figure 1. The design of the displacement pointer mechanism shall be such that an application of 500 microamperes of the proper polarity applied as a step input shall drive the displacement pointer up out of view. The application of 750 microamperes to drive the pointer out of view shall not damage the meter movement. This pointer shall perform as specified herein when connected to a 350 ohms signal source impedance and when connected in parallel with two each additional like meter movements or two each 1,000 ohms dummy loads.

3.4.7.4.1 Deflection. The direction of deflection for a given input shall be as specified in Table I.

3.4.7.4.2 Response. The displacement pointer response shall be linear in degrees with respect to current to within 7.5 percent of the proportionate full scale value. A deflection of 7/8 inch measured from the zero position of the scale to the center line of the pointer shall require 150 microamperes ± 5 percent.

3.4.7.4.3 Resistance. The resistance across the input terminals for the displacement pointer shall be 1000 ohms ± 3 percent at 25°C.

3.4.7.4.4 Damping. The displacement pointer mechanism shall be overdamped so as to preclude an overswing.

3.4.7.4.5 Response time. The response time of the displacement pointer shall be a minimum of 1.15 seconds and a maximum of 2 seconds. Response time is defined in 3.4.7.1.5.

3.4.7.5 Displacement pointer flag. The displacement pointer flag shall be in accordance with Figures 1 and 2. Its characteristics shall be the same as those specified for the vertical pointer flag alarm. Its terminal and polarization shall be as shown in Table I.

3.4.8 Rate of turn. The rate of turn display shall be in accordance with Figures 1, 3 and 4. The moving index shall be mounted as a meter movement which is pivoted at a point approximately 4-1/2 inches behind the pointer. The design shall be such that the pointer is as stable as practicable under rough air conditions. Within normal operating position, the index shall move left or right in agreement with the polarity of the direct current received from the remote rate of turn sensor. An external mechanical zero adjustment shall be provided. The scale shall be made to conform to the path of the index so that there shall be a minimum of parallax. The front surface of the scale and index shall be within 1/4 inch of the cover glass at zero displacement. The index deflection to either side of the center zero position shall be limited to 5/8 inch. At each of the fully deflected positions, a minimum of 1/2 of the index shall be in view when viewed at a horizontal distance of 2 feet from the center dot of the display.

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3.4.8.1 Deflections. The direction of deflection for a given input shall be as specified in Table I.

3.4.8.2 Response. The index response shall be linear in degree with respect to current to within 7.5 percent of the proportionate full scale value. A deflection of two index widths such that the index aligns with the appropriate outer scale mark shall require 1.0 ma ± 7.5 percent.

3.4.8.3 Resistance. The resistance across the rate of turn input terminals shall be 1,000 ohms ± 3 percent at 25°C.

3.4.8.4 Damping. The pointer mechanism shall be damped so that there shall be no more than 1-1/2 percent overshoot.

3.4.8.5 Response time. The response time of the rate of turn index shall be a maximum of 1/3 second. Response time is defined in 3.4.7.1.5.

3.4.9 Weight. The attitude director indicator including amplifier or AJB-3A adapter unit shall be so designed that when completely assembled the weight shall not exceed 9.0 pounds.

3.5 Electrical requirements.

3.5.1 Primary input power requirements. The indicator shall give the specified performance from power sources with characteristics as defined in MIL-STD-704 having limits as specified therein.

AC Power (Single Phase), 115V, 400 cps, Category "B", 30 VA

3.5.2 Undervoltage protection. The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within the specified limits.

3.5.3 Overload protection. Overload protection shall be provided in the indicator. All parts and circuits which are likely to carry an overload due to failures or poor adjustments shall have suitable protective devices or, preferably, shall be proportioned to withstand such overload without damage to the indicator. The indicator shall resume normal operation when the overload has been eliminated.

3.5.4 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the acquiring activity. Electronic devices and their applications shall be chosen in accordance with applicable MIL-STD-454 requirement as follows:

Requirement 30 - Semiconducter devices

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3.5.5 Electric connectors. The electric connectors used shall conform to MIL-C-26482 and MS18028. When the indicator is used with its removable amplifier, the pin connections of the indicator connector which mates with the amplifier connector shall be in accordance with Table II. The pin connections of the second indicator connector shall be in accordance with Table I. When the indicator is used with the AJB-3A system, the pin connections of the adapter unit connectors shall be in accordance with Figure 7.

3.5.6 Integral lighting. The indicator shall be provided with integral lighting in accordance with MIL-L-25467. The bank pointer may be back-lighted. All other display components shall be wedge-lighted. The lighting shall be subject to approval by the acquiring activity. (see 6.2.1)

3.5.7 Lamps. The indicator shall be provided with lamps as specified in MIL-I-25467.

3.5.8 Wiring. The internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the indicator case.

3.5.9 Power failure warning flag. A power failure warning flag shall be incorporated in the lower left of the face of the indicator as shown in Figures 1 and 2. The word "OFF" on the flag shall be marked in lusterless black material matching Color No. 37038 of FED-STD-595 on a background of lusterless white material matching Color No. 37875 of FED-STD-595. The "A" and "j" leads shall be brought out through the electrical connector in Table I. When either the ground connection or the power is removed, or both, the warning flag shall appear. Under night lighting the flag shall be front lighted. The meter and associated circuitry shall be designed to move the flag from view when 25 to 29 volts are applied.

3.6 Display. The display shall be securely mounted in such a manner that it shall not loosen or slip when the indicator is vibrated. If screws are used. They shall be so located that they shall not interfere with the proper location of any marking.

3.6.1 Visibility of display. The pointers, numerals, at least 1/16 inch of the shortest graduations and all other specified markings on the display shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the cover glass and whose small diameter is the aperture of the case.

3.6.2 Display marking. The display shall be marked as shown in Figures 1, 2, 3, 4, 5 and 6. The colors shall be in accordance with FED-STD-595. All markings shall be durable to withstand usage encountered in service. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch.

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3.6.3 Pointers. The pointers shall be as light as practicable, sufficiently rigid to prevent oscillation under vibration, and shall be firmly attached to the mechanism. The pointers shall be finished in accordance with Figure 3. The finish colors shall be in accordance with FED-STD-595.

3.7 Cover glass. The cover glass shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the indicator. The cover glass shall be 3/16 inch thick and shall be mounted in the flange by inserting it from within the case.

3.7.1 Anti-reflection coatings. A magnesium fluoride coating in accordance with MIL-C-675 shall be applied to each surface of the cover glass and of the lighting wedge. Coated surfaces shall not chip, scale, peel, flake, dissolve, discolor or otherwise be adversely affected by the environmental tests specified herein or in normal service use. The coating on the outer surface of the cover glass shall be capable of withstanding abrasion incident to normal service usage.

3.8 Performance. The indicator shall perform satisfactorily when subjected to the tests specified in section 4.

3.8.1 Reliability.

3.8.1.1 Operational stability. The indicator shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.8.1.2 Operation life. The indicator shall have total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.8.2 Reliability in Mean Time Between Failures (MTBF). For the test program under 4.4.3 the mean (operating) time between failures shall be 1,000 hours.

3.9 Interchangeability of reordered equipment. For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously, manufactured equipment, supplied or designated by the acquiring activity.

3.10 System operation. The indicator specified herein shall operate satisfactorily as part of the integrated flight instrumentation system meeting the requirements of MIL-S-23535.

3.11 Packaging. Each of the fully prepared shipping containers containing the indicator shall be examined to determine that the packaging, packing and marking conform to this specification. The list of defects, Table III, shall be used to enumerate the defects found.

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3.12 Identification of product.

3.12.1 Nameplate. A nameplate shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service. The nameplate shall be legibly marked in accordance with MIL-STD-130.

3.12.2 Manufacturer's part number. The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.13 Workmanship. The indicator, including all parts and accessories, shall be so constructed and finished that it shall be free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, plating, painting, riveting, machine screw assemblies and freedom of parts from burrs and sharp edges.

3.13.1 Dimensions. Each indicator shall be checked dimensionally to determine conformance to the dimensions in MS18028. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.13.2 Soldering. Soldering shall be performed in accordance with DOD-STD-1866.

3.13.3 Screw Threads. Screw threads, 0.0060 inch in diameter or larger, shall be in accordance with MIL-S-7742.

3.13.4 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Backlash shall be held to a minimum consistent with the application of the gear assembly.

3.13.5 Cleaning. The indicator shall be thoroughly cleaned and free from loose, spattered or excess solder, resin flash that may crumble, metal chips and other foreign matter, after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specifications shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or acutal, nor does it commit the Government to acceptance of defective material.

4.1.2 Test procedures. The procedures used for conducting all required tests shall be prepared by the contractor and submitted to the acquiring activity or authorized agent for review and approval. The right is reserved by the Government to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the acquiring activity. However, the right is reserved by the acquiring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.2 Classification of inspections. The inspection requirements specified herein are classfied as follows.

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of all of the tests of 4.6 of this specification and the first article inspection of MIL-S-23535 (see 6.5).

4.3.1 First article test samples. First article test samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for first article testing shall not have been previously tested except for the individual tests. The samples shall be forwarded to the testing laboratory designated by the acquisition activity (see 6.2.1).

4.3.1.1 First article test sample identification. The test samples shall be plainly identified by securely attached durable tags marked with the information listed below:

Sample for First Article Inspection
Remote Attitude-Director Indicator
Submitted by (Manufacturer's name, date)
for First Article Inspection in
accordance with MIL-I-23524A(AS)
under contract number .
MS - (Add proper dash number)
Manufacturer's Part Number.

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing all tests except sampling plan "B" tests which will be conducted at a laboratory designated by the

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acquiring activity. Quality conformance test, except for sampling plan "B", shall be under the supervision of the Government quality control representative. The contractor shall prepare and have available for review test reports showing detailed quantitative results for all tests required by this specification, signed by the authorized government quality control representative. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual test (see 4.4.1).

b. Sampling Plan "A" tests (see 4.4.2.1).

c. Sampling Plan "B" (see 4.4.2.2).

d. Reliability tests (see 4.4.3).

4.4.1 Individual tests. Each indicator submitted for acceptance shall be subjected to the Individual Tests listed below. These tests shall determine compliance with the requirements of design, construction, workmanship and operational accuracy. When the indicator fails any one of the tests at any point, the indicator shall be rejected and no further test shall be performed on the item. As a minimum, each indicator shall have passed the following tests:

a.	Examination of product - - - - - - - - - -	see 4.6.1
b.	Starting - - - - - - - - - - - - - - - - -	see 4.6.2
c.	Pitch, bank and azimuth zero - - - - - - -	see 4.6.3
d.	Sensitivity - - - - - - - - - - - - - - -	see 4.6.4
e.	Follow-up rate - - - - - - - - - - - - - -	see 4.6.5
f.	Follow-up accuracy - - - - - - - - - - - -	see 4.6.6
g.	Follow-up operation - - - - - - - - - - -	see 4.6.7
h.	Gimbal freedom - - - - - - - - - - - - - -	see 4.6.8
i.	Pitch trim - - - - - - - - - - - - - - - -	see 4.6.9
j.	Zeroing adjustment - - - - - - - - - - - -	see 4.6.10
k.	Leakage - - - - - - - - - - - - - - - - -	see 4.6.11
l.	Power flag warning indicator - - - - - - -	see 4.6.12
m.	Inclinometer sensitivity - - - - - - - - -	see 4.6.13
n.	Inclinometer zero position - - - - - - - -	see 4.6.14
o.	Inclinometer friction - - - - - - - - - -	see 4.6.15
p.	Inclinometer visibility - - - - - - - - -	see 4.6.16
q.	Inclinometer filling - - - - - - - - - - -	see 4.6.17
r.	Pointer response - - - - - - - - - - - - -	see 4.6.18
s.	Pointers balance - - - - - - - - - - - - -	see 4.6.19
t.	Pointer flag alarms - - - - - - - - - - - -	see 4.6.20
u.	Lighting test - - - - - - - - - - - - - - -	see 4.6.21

4.4.2 Sampling plan tests. The sampling plan tests shall consist of sampling plan A and sampling plan B tests. Indicators selected for sampling tests shall first have passed the individual tests. Indicators which have been subjected to the sampling plan A Tests shall not be delivered on contract until they have been refurbished and resubmitted for all the Individual Tests. Indicators which have been subjected to the sampling plan B Tests shall not be delivered on contract.

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4.4.2.1 Sampling plan A sample selection. Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of indicators offered for acceptance	Quantity selected for testing
First 15	1 *
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	

* When sampling plan B is invoked the quantity is 0. When sampling plan B is omitted the quantity is 1.

When a defective indicator is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated (one from next 50, etc.).

4.4.2.1.1 Sampling Plan A tests. Each indicator selected for sampling plan A testing shall be subjected to the tests listed below:

a.	Inclinometer damping at room temperature	see 4.6.22
b.	Pointers response time	see 4.6.23
c.	Power consumption	see 4.6.24
d.	Low voltage	see 4.6.25
e.	Magnetic effect	see 4.6.26
f.	Acceleration sampling plan A	see 4.6.27
g.	Dielectric strength	see 4.6.28
h.	Vibration error	see 4.6.29
i.	Voltage and frequency variation	see 4.6.30
j.	Low temperature operation	see 4.6.31
k.	High temperature operation	see 4.6.32
l.	Fogging	see 4.6.32.1
m.	Slip indicator leak	see 4.6.32.2

4.4.2.2 Sampling plan B instructions. Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These indicators shall be forwarded to a laboratory designated by the acquisition activity (see 6.2.1). Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Indicator, Attitude-Director; Remote
Submitted by (Manufacturer's name, date)
for quality conformance sampling plan B
testing, in accordance with Contract .
Manufacturer's part number .

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4.4.2.2.1 Sampling Plan B Approval. Approval of Sampling Plan B indicator shall be by the acquiring activity upon satisfactory completion of the designated tests. Any material, design or performance defect made evident during these tests shall be corrected by the Contractor to the satisfaction of the acquiring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of indicators under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Sampling plan B tests. Each indicator selected for sampling plan B testing shall be subjected to the tests listed below:

a.	Sampling plan A tests	see 4.4.2.1
b.	Electromagnetic interference	see 4.6.33
c.	High temperature exposure	see 4.6.34.1.1
d.	High altitude, low temperature	see 4.6.34.1.2
e.	Temperature shock	see 4.6.34.1.3
f.	Acceleration Sampling Plan B	see 4.6.34.1.4
g.	Vibration endurance	see 4.6.34.1.5
h.	Life	see 4.6.34.1.6
i.	Humidity	see 4.6.34.1.7
j.	Fungus resistance	see 4.6.34.1.8
k.	Salt fog	see 4.6.34.1.9
l.	Impact shock	see 4.6.34.1.10

4.4.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. The indicators selected for reliability assurance tests shall have first passed the Individual Tests.

4.4.3.1 Reliability procedure. For the reliability qualification plan, as outlined in MIL-STD-781 shall be used.

4.4.3.2 Test level. The test shall be as follows: Except for the initial temperature of +71°C (+160°F) followed by -54°C (-65°F) the high temperature shall be +55°C (+130°F) and the low temperature shall be -40°C (-40°F). The initial temperature cycle shall be conducted at the start of each test and on each unit thereafter upon repair or replacement. The length of each cycle shall not exceed 24 hours nor shall either of the high or low temperature conditions be less than 4 hours. The unit shall be vibrated at 2.0g at a non-resonant frequency between 20 and 60 cps for ten minutes at each extreme temperature.

4.4.3.3 Duty cycle. Upon receipt of the indicator, following operational test (or debugging), it shall be placed in a chamber at +71°C (+160°F) and operated for 4 hours using nominal voltage. Signals shall be applied to exercise the indicator ball at least ± 5 degrees in all 3 axes and the pointers at least ± 1/2 full scale at a frequency between 5 and 7 cycles per minute during the test. The warning flags shall be driven out of view.

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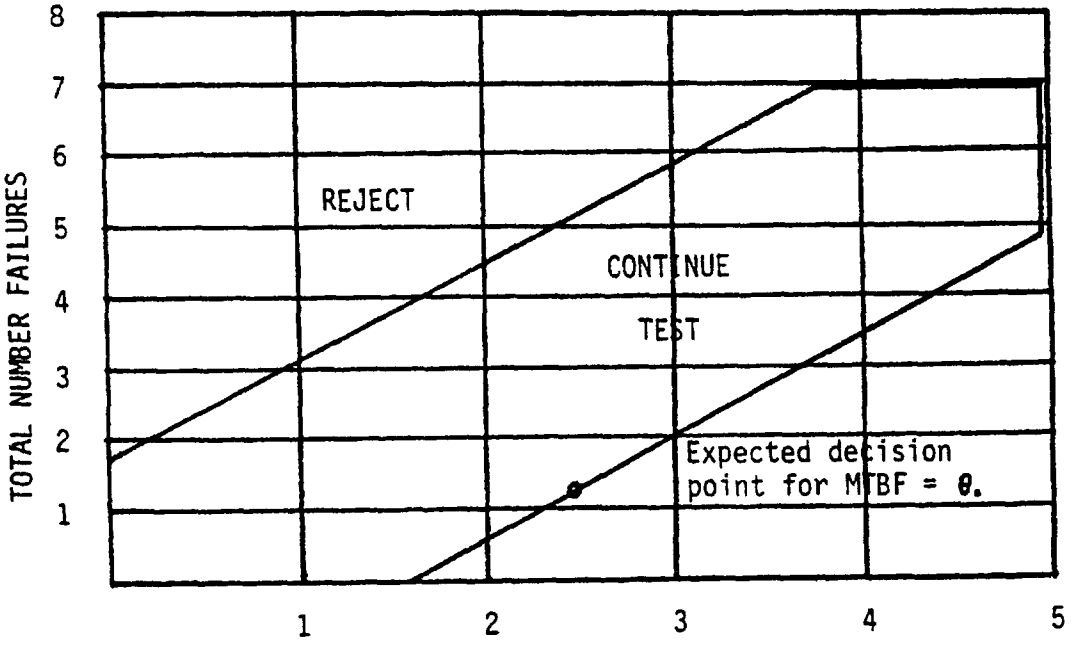
The indicator shall then be tested at -54°C (-65°F) after a 4 hour soak at -54°C (-65°F). The test then continues as follows:

The indicator shall be operated for 2-1/2 hours at nominal voltage and shall be de-energized for 1/2 hour. This duty cycle shall be conducted eight times each day. Time included toward the reliability test shall be the sum of all 2-1/2 hour running periods.

Following the hot and cold tests of the initial temperature cycle (see 4.4.3.3) the test shall continue at -40°C (-40°F) to +55°C (+130°F). At all times when power is applied during the reliability test, including hot and cold, the indicator 3 axes ball and the pointers shall be exercised at between 5 and 7 cycles per minute and at displacement of at least ± 5 degrees in each axis for the ball and ± 1/2 scale indication for the pointers and the warning flags shall be driven out of view.

4.4.3.4 Accept/reject criteria. The following graph shall be used to determine the accept/reject criteria for the reliability qualification plan. The test must be completed within four months of the date of the delivery and if a 1,000 hour MTBF has not been demonstrated, acceptance of the indicators will be withheld until corrective action is taken and its effectiveness is demonstrated. If a 1,000 hour MTBF has not been demonstrated, a repeat of the test must be initiated within thirty days on new equipments selected by the Government quality assurance representative.

Accept-reject criteria	
Decision risks	20 percent
Discrimination ratio	2.0 : 1



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No. of failures	Total Test Time * Reject (equal or less)	Accept (equal or more)
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

* Total test time is total unit hours of "equipment on" time and is expressed in multiples of the specified MTBF.

4.4.3.5 Performance characteristics to be measured. Tests necessary to determine failures in performance characteristics (listed below) shall be conducted at least once every 24 hours. If operation is extended over a weekend, the tests may be conducted on the day prior to and the day following the weekend in lieu of the 24-hour tests for the weekend. A failure shall be defined to be any of the following conditions:

a. Any failure which prevents the presentation of accurate roll, pitch, azimuth, etc. information.

b. Any intermittent condition which causes jumping in the roll, pitch, azimuth, etc. indication. Disqualify jumping as specified herein.

c. Any condition which prevents the pointers from presenting an accurate indication as specified herein.

d. Any failure of the warning flags to operate properly.

4.4.3.6 Failure criteria. In addition to the requirements of MIL-STD-781, the following requirement shall be used to determine when a failure has occurred during the test:

a. When performance characteristics fall below the acceptance requirements (see 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused other parts to fail.

4.4.3.7 Preventive maintenance. During the period of the tests, no preventive maintenance shall be performed on any of the items. Simple adjustments, normally performed by the operator on such equipment shall be permitted. However, a log of all such adjustments shall be kept. The acquiring activity may weigh these adjustments and if it feels justified assign excessive ones as failures.

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4.5 Test conditions.

4.5.1 Standard conditions. Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25°C ± 5°C
Pressure	Normal atmospheric (approximately 29.92 inches Hg)
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input voltage	115 ± 1.0 volts AC

4.5.2 Test readings. Except where otherwise specified, the indicator shall be vibrated before a test reading is taken. An electrical vibrator set at 30, 60 or 120 cycles per second with maximum amplitude of 0.002 inch when attached to the test item, shall be used.

4.5.3 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position with the dial vertical.

4.5.4 Lighting. Unless otherwise specified, the indicator lighting system shall be energized and shall operate satisfactorily during and after all tests except humidity, fungus resistance and salt fog.

4.5.5 Standard test transmitter. The standard test transmitter shall be an Eclipse-Pioneer Type AY201-1, or equivalent, with minimum and known errors. These errors shall be subtracted from those of the indicator under test. The transmitter shall have an accurate dial equipped with a ± 1 degree vernier and readable to 0.1 degree. The dial shall be graduated through 360 degrees and shall be capable of continuous rotation. The transmitter rotors shall be excited from the same power supply voltage which operates the indicator. Excitation shall be applied so that the voltage from rotor lead "C" to rotor lead "H" is in phase with the voltage from pin A to pin B of the indicator with lead "C" and pin A common as shown in Table I. The transmitter shall be calibrated and the rotor and stator leads identified as follows:

a. Designate the rotor leads as "H" and "C", "C" to be grounded.

b. Designate one stator lead as "Z" and connect it to "C".

c. Apply 26V, 400 cps across "H" and "C" and measure the voltage across the two remaining stator leads and the voltage across "H" and one of the stator leads.

d. Turn the rotor until the voltage across "H" and the one stator lead is approximately maximum (about 35 volts) and the voltage across the two stator leads is exactly minimum. Lock the dial to the rotor with the dial exactly on zero.

e. Turn the dial through increasing numbers and measure the voltage across the two undesignated stator leads and "Z". The voltage across "XZ" will increase before decreasing and the voltage across "YZ" will decrease before increasing.

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4.5.6 Properly connected. When the indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture which is providing a lighting circuit voltage variable from 5 volts (AC or DC) to approximately zero volts, synchro signals in accordance with 3.4.5.1 through 3.4.5.7 variable current DC signals 1,000 ohm source and with reversible polarity to the meter movements. AC power as required, and the pin J is jumped to pin A (Table I) through a closed switch.

4.5.7 Definition of indicator. Unless otherwise specified in a particular test, the word "indicator" as used in Section 4 of this specification shall be construed to mean the indicator proper together with an amplifier or adapter unit, as applicable.

4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be examined externally to determine conformance to the applicable requirements in packaging (see 3.11), design and construction (see 3.4), identification of product (see 3.12) and workmanship (see 3.13).

4.6.2 Starting. The indicator shall be properly connected and when power is applied the power warning flag shall disappear and the indicator shall be ready for immediate operation.

4.6.3 Pitch, bank and azimuth zero. The standard test transmitters in the pitch and roll axes shall be set to the 180 degrees dial reading and the standard test transmitter in the azimuth axis set to the zero degree dial reading. Power shall be applied and the indicator shall follow-up to the zero indications in pitch, bank and azimuth within ± 1/2 degree from any random indication in any axis. This test shall be repeated for any initial indication at the discretion of the government quality assurance representative. The pitch and roll indications shall be adjusted to zero by means of the zeroing adjustments (see 3.4.5.8).

4.6.4 Sensitivity. The test transmitter for the roll axis shall be moved 1/4 degree in right roll and returned to the original position, then 1/4 degree in left roll and returned to the original position. The test transmitter for the pitch axis shall be moved 1/4 degree in dive and returned to the original position, then 1/4 degree in climb and returned to the original position. The test transmitter for the azimuth axis shall be moved 1/4 degree left turn and returned to its original position, then 1/4 degree right turn and returned to its original position. There shall be a smooth perceptible movement of the indicator bank index, horizon line and azimuth index for each movement of the appropriate test transmitter.

4.6.5 Follow-up rate. The indicator shall be properly connected to a remote signal source or to a standard test transmitter. The follow-up rates in pitch, roll and azimuth shall be determined. The rate in the roll and azimuth axis shall be not less than 300 degrees per second. The rate in the pitch axis shall be not less than 90 degrees per second.

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4.6.6 Follow-up accuracy. The standard test transmitter in the roll axis shall be displaced to at least 10 random positions other than the 180 degrees dial reading. The indicator shall position itself with the rotor of the standard test transmitter within ± 1/2 degree for each position within ± 30 degrees indication and within one degree at all other indications. The standard test transmitter in the pitch axis shall be displaced to at least 10 random positions other than the 180 degrees dial reading. The indicator shall position itself with the standard test transmitter within ± 1/2 degree for each position within ± 30 degrees indication and within ± 1 degree for all other indications. The standard test transmitter in the azimuth axis shall be displaced to at least 10 random positions other than the 0 degree dial reading. The indicator shall position itself with the standard test transmitter within ± 1/2 degree for each position within ± 30 degrees indication and within one degree for all other indications.

4.6.7 Follow-up operation. Suitable test equipment shall be connected to the appropriate test pins of the indicator to observe and record any hunting, jumping, lag, or overshooting of the indication when the indicator is made to follow the standard transmitter at various constant rates in the roll, pitch and azimuth axes. This must be possible with the indicator sealed. The recorded data shall meet the following requirements. This data shall be recorded during the final Individual Tests following the completion of all other tests. While performing the individual tests at other times the indicator may be checked visually for noticeable hunting ,jumping, lag or overshooting.

4.6.7.1 Indication lag. Indication shall not lag behind the shaft of the remote input transmitting synchro by more than the amount shown in the curve on Figure 8. The Indicator shall be tested at three different rates (including the maximum rate specified in 4.6.5) for each servo. The servos may be checked at any rate within the maximum rate, at the discretion of the Government quality assurance representative. A 25 percent increase in indication lag shall be permitted during temperature extremes and low voltage tests.

4.6.7.2 Hunting and jumping. There shall be no noticeable hunting or jumping when the standard transmitter shaft is not rotating. The inspector shall select five indicators at random which shall be operated over the full range of follow up speeds up to 300 degrees per second for roll and azimuth and 90 degrees per second for pitch. The rates chosen for testing hunting or jumping shall be those which produce the most erratic motion. Either the peak to peak amplitude of the hunting or jumping, or both, at the speed determined shall not exceed one degree.

4.6.7.3 Overshooting. Overshooting upon the completion of pitch or roll maneuvers shall be held to a practical minimum and shall not exceed one overswing with a maximum amplitude of 3 degrees, with a step input equal to the maximum follow-up lag.

4.6.8 Gimbal freedom. The indicator shall be properly connected and the remote signal source or standard test transmitter moved 360 degrees in roll, pitch and azimuth. The indicator shall follow smoothly without sticking and overshooting or hunting greater in magnitude than that outlined in 4.6.7 through 4.6.7.3.

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4.6.9 Pitch trim. The pitch trim knob shall be adjusted to its limits and the deflection of the horizon line observed. The deflection shall be from 5 to 10 degrees down and from 10 to 20 degrees up from the level flight indications. The indicator shall meet the pitch trim fading requirements of 3.4.4.4.

4.6.10 Zero adjustments. The indicator shall be properly connected to the standard test transmitters, each set to electrical zero. The roll zeroing adjustment shall be moved to its extremes and the deflection of the bank index shall be not less than ± 3 degrees from the zero position. The pitch test transmitter shall be rotated 3 degrees first in climb, then 3 degrees in dive. In each case the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zeroed. The pitch trim knob shall be tightened again to the shaft at the zero position. In each case the requirements of 4.6.9 shall be met.

4.6.11 Leakage. Each indicator shall be tested for case leakage with either a mass spectrometer or by an equally sensitive leak test. The gas or mixture of gases in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding one micron cubic foot per hour (10^{-5} cubic centimeters per second).

4.6.12 Power flag warning indicator. The indicator shall be properly connected, except that no power shall be applied. Power shall be applied and the flag shall disappear. The power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from pin J to A shall be disconnected. The flag shall appear.

4.6.13 Inclinometer sensitivity. With the dial vertical, the indicator shall be rotated about the roll axis to the right until the ball is just short of its limit. The angle of rotation shall be 10 ± 2 degrees. The test shall be repeated rotating the indicator to the left. The same tolerance shall apply. The ball shall not stick at any position.

4.6.14 Inclinometer zero position. The indicator shall be positioned with the upper edge of the mounting bezel horizontal, and the indicator shall be gently tapped. The ball shall rest at the zero index mark of the slip indicator within 1/32 inch.

4.6.15 Inclinometer friction. The indicator shall be slowly tipped to either side in the vertical plane of the dial. The ball shall roll smoothly when the indicator is gently tapped.

4.6.16 Inclinometer visibility. The indicator shall be so tipped that the ball is at rest at either end of the tube. Not less than 2/3 of the ball shall be visible when the ball is viewed from the position specified in 3.4.6.2.

4.6.17 Inclinometer filling. The indicator shall be so tipped that all the air in the tube shall be trapped in the expansion chamber end of the tube. The face of the indicator shall be in a vertical plane with the line joining the centers of the two lower mounting holes in a horizontal plane. No part of the air bubbles shall be visible when the indicator is viewed from the position specified in 3.4.6.2.

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4.6.18 Pointers response. The response of the vertical pointer, the horizontal pointer, the displacement pointer, and the rate of turn pointer shall be measured and shall be in compliance with the requirements of 3.4.7.1.2, 3.4.7.3.2, 3.4.7.4.2 and 3.4.8.2, respectively.

4.6.19 Pointers balance. With the indicator in a normal upright position and the pointer meter mechanisms de-energized, the vertical and horizontal pointers shall each be adjusted by means of the zero correctors to bisect the center for the indication. The pointers shall not be deflected by more than 1/64 inch as the indicator is rotated 360 degrees in roll and pitch. The 1/64 inch limit shall also apply to the rate of turn pointer while being rotated in roll and pitch. The displacement pointer shall not deflect more than 1/64 inch when the indicator is rotated in roll nor more than 1/32 inch when the indicator is rotated in pitch.

4.6.20 Pointer flag alarms. The response of all pointer flag alarms shall be measured and shall be in compliance with the detail requirements of 3.4.7.2.2.

4.6.21 Lighting test. The lighting circuits of the indicator shall be properly connected. The indicator shall be tested in accordance with MIL-L-25467 and shall comply with the requirements of that specification. A prime lighting standard approved by the Naval Air Systems Command (see 6.2.1) shall be used for conducting the lighting tests.

4.6.22 Inclinometer damping at room temperature. The plane of the face of the indicator shall be placed in a vertical position with the upper flange inclined to the horizontal at an angle of 12 degrees. The indicator shall be tipped suddenly through a vertical angle of 24 degrees in order that the ball shall roll to the opposite end of the tube. The time of roll of the ball from the zero mark of the indicator to the end of the tube shall be between 0.15 and 1.0 second.

4.6.23 Pointer response time. The response time of the indicating pointers shall be measured and shall be in accordance with the requirements of 3.4.7.1.5, 3.4.7.3.5, 3.4.7.4.5 and 3.4.8.5, respectively. For this test the indicator shall be shunted externally with a resistance of 350 ohms, to allow for the shunting effect of another indicator in parallel with the usual source. The voltage source shall be steady and shall be applied to the above circuit through at least 50,000 ohms. At least 5 seconds shall be allowed in this test in order to establish the calibration of the 100 percent mark.

4.6.24 Power consumption. The power consumption of the indicator shall be measured with suitable meters. The AC consumption shall not exceed 30 volt-amperes. The power factor shall be maintained between 0.70 and 1.00. Lighting power shall not be included.

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4.6.25 Low voltage. The indicator shall be properly connected. The AC voltage shall be reduced to 103 volts and 380 cps. The system shall meet the requirements specified in 4.6.2 through 4.6.10 and 4.6.12.

4.6.26 Magnetic effect. This test shall be performed first with no power applied to the indicator and then shall be repeated with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 12 inches from the center of a free magnet, approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ± 0.01 oersted, and the maximum deflection of the magnet shall not exceed 10 degrees. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

4.6.27 Acceleration sampling plan A test. The indicator shall be properly connected and power applied to the attitude circuit. The indicator shall be accelerated along its normal vertical axis with a force of 10 g. The attitude display shall not vary by more than one degree in pitch, roll or azimuth from the initial indications and the pointers shall not vary by more than 5/64 inches from the zero positions when the meter pivot axes are parallel to the spin axis of the centrifuge. This test shall be conducted with the attitude set at zero pitch, roll and yaw and with each axis displaced 90 degrees.

4.6.28 Dielectric strength. With the indicator properly connected to the amplifier unit, a potential of 200 volts direct current shall be applied between isolated pins of the connector and between pins and the indicator case for a period of 5 seconds. There shall be no breakdown of insulation.

4.6.29 Vibration functional. The indicator shall be rigidly attached to the vibration machine using the mounting holes. The indicator shall be connected and energized as described in paragraph 4.6.3. The indicator shall be operated by the standard test transmitters during the vibration test. The vibration test shall be a random vibration spectrum, (see Figure 9, Level A). The indicator shall be subjected to and shall meet the requirements of the individual tests following the functional vibration. No condition of applied vibration shall cause vibration errors involving pointer oscillation or variation in excess of the tolerance outlined below.

4.6.29.1 Roll, pitch, azimuth and director pointers. The maximum total amplitude of roll, pitch, azimuth, horizontal pointer, vertical pointer and displacement pointer (hereinafter referred to as "the pointers") oscillation, throughout the frequency range shall be noted and shall not exceed 1-1/2 degree in roll, pitch, and azimuth or 1-1/2 of the width of the respective pointer.

4.6.29.2 Roll, pitch, azimuth and pointer variation. The maximum variation during vibration of the roll, pitch, azimuth and pointers, from the original indication before vibration throughout the frequency range, shall be noted and shall not exceed 1/2 degree in roll, pitch and azimuth or 1/2 of the width of the respective pointer.

4.6.29.3 Oscillation and variation. Oscillation and variation tests may be conducted at any indication at the discretion of the inspector.

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4.6.29.4 Inclinometer accuracy. The slip indicator reading shall not differ at any time by more than 1/16 inch from the reading obtained when the indicator is at rest.

4.6.30 Voltage and frequency variation. The indicator shall be properly connected to standard test transmitters in the pitch, roll and azimuth axes. The transmitters shall be set to drive the indications in pitch, roll and azimuth to the zero position. The input voltage and frequency shall be varied within the following limits in any combination.

Frequency	380 to 420 cps
AC voltage	103 to 127 volts

The horizon line shall not shift more than ± 1/16 inch from normal null position. The bank and azimuth index shall not shift more than 1/4 degree from normal null position.

4.6.31 Low temperature operation. The indicator shall be properly connected, except that no power shall be applied, and placed in a chamber at a temperature of -54 ± 2°C for a period of 4 hours. At the end of this period, and while the indicator is at the reduced temperature, power shall be applied for a period of one minute. At the end of this period but still at the reduced temperature, the indicator shall meet the requirements specified in 4.6.7.1, 4.6.8, 4.6.9, 4.6.12 and 4.6.22, except that the time of roll in 4.6.22 shall be 4 seconds maximum and the filling liquid of the inclinometer shall not become cloudy nor shall flakes appear. Meter movements shall be checked only for operation, for evidence of sticking or binding and to ascertain that they have not shifted from their null positions by more than 1/64 of an inch.

4.6.32 High temperature operation The indicator shall be properly connected and placed in a chamber at a temperature of +71 ± 2°C and operated with power applied for a period of 4 hours. At the end of this period and while at the high temperature, the indicator shall meet the requirements specified above for low temperature operation, except that in 4.6.22 the normal maximum time of roll shall be 1.0 second.

4.6.32.1 Fogging. The indicator shall be energized for a minimum of 1 hour in a 71 ± 2°C controlled ambient. After this period and while at the specified temperature, an ice cube shall be rubbed on the indicator glass face for 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection. (see 3.4.2.3)

4.6.32.2 Slip indicator leak. The size of the air bubble in the tube shall be determined with the indicator at room temperature prior to the high temperature test. The size of the air bubble shall then be determined with the indicator at room temperature following the test. There shall be no appreciable change in the size of the air bubble before and after the indicator has been subjected to the high temperature.

4.6.33 Electromagnetic interference. The electrical system of the indicator shall be tested for conducted and radiated electromagnetic interference in accordance with MIL-STD-461.

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4.6.34 Performance record. Prior to conducting any of the following tests, the indicator shall be subjected to a comprehensive operational test under standard conditions (see 4.5.1) and a record made of all data necessary to determine compliance with this specification. This data shall provide the criteria for checking satisfactory performance of the indicator either during or at the conclusion of the test, or both, as required. The indicator shall be operated prior to test to insure that no malfunction exists.

4.6.34.1 Evaluation of the indicator. When so directed in Individual Test procedures, the indicator undergoing test shall be operated to permit performance data to be obtained, or inspected for evidence of deterioration. The performance data of the indicator under test conditions of following test shall be satisfactorily comparable to that obtained in compliance with 4.6.34. Any deteriortaion observed shall not exceed that defined in 4.7.

4.6.34.1.1 High temperature exposure. The indicator shall be properly connected except that no power shall be applied and shall be placed within the test chamber. The internal temperature of the chamber shall then be raised to 71°C (160°F). The indicator shall be exposed to that temperature for a period of 48 hours. A relative humidity of not more than 15 percent shall be maintained in the test chamber throughout the exposure period. After the indicator has returned to room temperature, it shall meet all the requirements of all Individual Tests (see 4.4.1 and 4.6.34.1).

4.6.34.1.2 High altitude, low temperature. The indicator and amplifier shall be properly connected, except that no power shall be applied. The indicator and amiplifier shall be placed within the test chamber and chamber cooled to and maintained at a temperature of -62°C (-80°F) for a period of 72 hours, at which time the indicator and amplifier shall be inspected in accordance with the requirements of the Individual Tests (see 4.4.1). The temperature of the chamber shall then be raised to -54°C (-65°F) and maintained for an additional 24-hour period or until stabilization of equipment temperature is reached, whichever is the longer. At the conclusion of this exposure period and while still at this temperature, the pressure in the chamber shall be reduced to the equivalent of 90,000 ± 500 feet and power appplied. There shall be no binding or sticking at any possible indication. After the indicator and the amplifier has returned to room temperature, it shall meet the requirements of all Individual Tests (see 4.4.1 and 4.6.3.4.1).

4.6.34.1.3 Temperature shock. The indicator shall be tested in accordance with temperature shock test of MIL-T-5422, except that the high temperature and the low temperature extremes shall be 71°C (160°F) and -40°C (-40°F) accordingly. After completion of the test, the indicator shall meet the requirements of all individual tests (see 4.4.1 and 4.6.34.1).

4.6.34.1.4 Acceleration sampling plan B test. The indicator, not operating, shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of 2 axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20 g in each of the above axes for a period of 1 minute. At the end of this acceleration, the indicator shall meet the requirements of the Individual Tests (see 4.4.1). No damage to the indicator shall result from this test.

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4.6.34.1.5 Vibration endurance. The indicator shall be rigidly attached to the vibration machine using the mounting holes. The vibration test shall be subjected to a random vibration spectrum, (see figure 9, Level B). The indicator shall be subjected to and meet the requirements of the individual tests.

4.6.34.1.6 Life. The indicator shall be properly connected and operated such that the roll, pitch and azimuth axes of the attitude display are oscillated through a minimum of ± 30 degrees, and all pointers are driven through their complete indicating range (none of the pointers driven out of view) 5 to 7 times per minute. The power failure warning flag and two pointer flags shall be de-energized momentarily every 30 minutes. The displacement pointer, and vertical pointer and horizontal pointer shall be driven out of view every 30 minutes. Five volts AC or DC shall be applied continuously to the lighting circuit. The life test shall last for 1,000 hours. The test may be interrupted at 250 hour intervals to determine compliance with part or all of the Individual Tests (see 4.4.1) There shall be no failure during the test. After 1,000 hours operation, the indicator shall meet the requirements of the Individual Tests (see 4.4.1). The testing agency may run this life test on three (3) hour cycle, 2-1/2 hours on and 1/2 hour off, in which case hours of life shall be the sum of the periods during which the indicator is energized.

4.6.34.1.7 Humidity. The indicator shall be tested in accordance with humidity test of MIL-T-5422. Upon completion of the test, the indicator shall meet the requirements of 4.6.4 through 4.6.9. There shall be no evidence of corrosion or rusting of the internal or external parts of the indicator assembly which would adversely affect subsequent operation.

4.6.34.1.8 Fungus resistance. The indicator shall be tested in accordance with the procedure listed below:

Procedure - Four groups of fungi are listed below and one species of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at a temperature between 22°C and 32°C (72°F and 89°F) shall be utilized. Approximately 10 ml of the solution shall then be introduced directly into a tube culture of the fungus and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an innoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions from the four types of fungi shall be mixed together to provide a composite suspension. Actively growing cultures between 7 to 21 days old after initial innoculation shall be used for the preparation of the spore suspension. After preparation, the spore suspension shall not be kept for more than a 24-hour period at temperatures from 22°C to 32°C (72°F to 89°F) or not more than 48

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hours at temperatures from 2°C to 7°C (35°F to 45°F). The equipment, including applicable external connections, shall be placed in an incubation chamber, maintaining an internal temperature of 30°± 2°C (86 degrees ± 3.6°F) and a relative humidity of 95 ± 5 percent, and sprayed with the suspension of mixed spores. The test period shall be 28 days. At the end of the test, the parts shall be visually examined in accordance with 4.6.34.1. Upon completion of the test, the indicator shall meet the requirements of 4.6.4 through 4.6.7. There shall be no deterioration which adversely affect subsequent operation and fungus growth shall not be supported.

4.6.34.1.8.1 Organism.

Group I	-	Chaetomium globosum	6205
		Myrothecium verrucaria	9095
Group II	-	Memenoniella echinata	9597
		Aspergillus niger	6275
Group III	-	Aspergillus flavus	10836
		Aspergillus terreus	10690
Group IV	-	Penicillium citrinum	9849
		Penicillium ochrochloron	9112

4.6.34.1.8.2 Stock culture designation and source. When ordering cultures, request the cultures by name and serial number. The serial numbers of the cultures remain the same regardless of source.

Source - American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852.

4.6.34.1.9 Salt fog. The indicator shall be properly connected, except that no power shall be applied. The indicator shall then be tested in accordance with salt fog test of MIL-T-5422. At the end of the 48 hour period, the indicator shall meet the requirements of 4.6.4 through 4.6.7., 4.6.9 and 4.6.10.

4.6.34.1.10 Impact shock. The indicator shall be rigidly attaced to the shock machine using the mounting holes. The indicator shall be connected and energized as described in paragraph 4.6.3. The indicator shall be operated by the standard transmitters during the shock test. Three shocks in each direction shall be applied along the three matually perpendicular axes of the indicator (total of 18 shocks). The shock pulse shall be one-half sinusoidal with a peak value of 15 + 2g and a duration of 11 milliseconds. The indicator shall be subjected to and meet the requirements of the individual tests following completion of the shock test. A crash safety test shall be performed to test the integrity of the mounting. The indicator or dummy load shall be attached by its normal mounting holes. The indicator or dummy load shall be subjected to two shocks in each direction along the three mutually perpendicular axes (total of 12 shocks). The shock pulse shall be one-half sinusoidal with peak value of 30 + 5G and a duration of 11 milliseconds. There shall be no failure of the mounting attachment and the indicator or dummy load shall remain in plce and not create a hazard. Bending and distoration shall be permitted.

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4.7 Criteria for failure. Deterioration or change in performance of any components which could in any manner prevent the indicator from meeting functional, maintenance and service requirements during service life shall provide reason to consider the indicator as having failed to comply with the conditions of the test to which it was subjected.

4.7.1 Rejection and retest. The indicator which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance and the indicator shall meet all the requirements specified herein. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defect found in the original shall be furnished to the government inspector.

4.8 Maintainability demonstration. Unless otherwise specified (see 6.2), maintainability demonstration test shall be conducted in accordance with Military Standard MIL-STD-471. The tests shall be selected in advance and shall be approved by the acquiring activity. The indicator shall have a median value of corrective maintance tasks (Mct) of 6.0 hours when Test Method 1, Test Plan A, of MIL-STD-471 is used. The max Mct shall be no longer than 8.0 hours. Maintainability demonstration shall be performed on a minimum of three selected repairs.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B or Industrial, as specified (see 6.2.1).

5.1.1 Level A. The indicator shall be preserved in accordance with Method IIC of MIL-P-116. The indicator shall be cushioned by envelopment with flexible, cellular plastic film material (PPP-C-795, PPP-C-1842, PPP-C-1797 or PPP-C-1752) and taped. The cushioned indicator shall be inserted, with four units of desiccant, into a bag fabricated of MIl-B-131 barrier material and the bag heat sealed. The bagged item shall be placed into a handling case, National Stock Number (NSN) 8145-01-016-3455 or blocked/braced into a single wall, corrugated fiberboard box (PPP-B-636, RSC, Class WR) of maximum exterior dimensions, 13x9x8 inches (330x229x203mm). The handling case or box shall then be installed on the shock-mounted platform of plastic shipping and storage container, NSN 8145-01-016-3451. Both the handling case and shipping/storage container are stocked by the Navy Aviation Supply Office (ASO), Philadelphia, PA, and are cited in MIL-STD-794 Appendix G. If the outer shipping/storage container is not available, the packaging procedures shall be developed in accordance with MIl-G-81559.

5.1.2 Level B. The indicator shall be preserved and packaged as described in 5.1.1, except that desiccant may be omitted.

5.1.3 Industrial. Preservation shall be in accordance with ASTM-D-3951.

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5.2 Packing. Packing shall be Leval A, B or Industrial as specified (see 6.2.1). The reusable shipping/storage container cited in 5.1.1 satisfies all levels of packing. If alternate packaging is used per MIL-G-81559, the specified level of packing protection shall be compatible with the criteria of MIL-STD-794 exterior shipping containers.

5.2.1 Industrial. The indicator, preserved as specified in 5.1, shall be packed in accordance with ASTM-D-3951.

5.3 Marking. The marking of the barrier bag and exterior shipping container shall be in accordance with MIL-STD-129 and the following precautionary markings shall appear on two opposite sides of each exterior shipping container:

DELICATE INSTRUMENTS HANDLE WITH CARE.

In addition, the following shall be bodly applied to each interior container:

CONTENTS EXTREMELY DELICATE - HANDLE WITH CARE
HANDLING CASE ONLY
DO NOT USE AS SHIPPING CONTAINER

6. NOTES

6.1 Intended Use. The remote attitude director indicator covered by this specification is intended for use in high performance aircraft as part of an integrated panel, and also as part of the AJB-3A Bomb Release Computing Set. The indicator will be controlled by a vertical reference Control and heading reference system providing transmitting synchro pitch, roll and azimuth output signals and a flight director computer providing outputs to the horizontal and vertical director pointers. The rate of turn indicator shall be controlled by a remote sensor.

6.2 Ordering Data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. The quantity and MS part number of the instrument desired.

c. Approval of anodic treatment when used on magnesium parts (see 3.3.3.3).

d. Approval of fungicidal treatment when used (see 3.3.3.6).

e. Approval of special tool or jig when used for sealing the case (see 3.4.2.1).

f. Approval of a mixture of gases as the filling medium when used (see 3.4.2.2).

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g. Approval of the integral lighting of the indicator (see 3.5.6).

h. Name and address of first article inspection laboratory (see 6.2.1).

i. Name and address of laboratory for Sampling Plan B tests (see 4.4.2.2).

j. Approval of prime lighting standard by Naval Air Systems Command (see 4.6.21).

k. Approval of not conducting the maintainability demonstration test (see 4.8).

l. Specify levels of preservation and packing (see 5.1 and 5.2).

6.3 Drawing requirements. The attention of contractors is invited to the fact that Government contracts for equipment, whether directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of DOD-STD-100 and DOD-D-1000.

6.4 First article. When first article inspection is required, the indicator shall be tested and shall be a first article sample. The first article shall consist of the samples specified in 4.3. The contracting officer shall include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.4.1 First article provision. The manufacture of indicators on contract shall not commence until the samples submitted are pronounced satisfactory by the acquiring activity. When a contractor is in continuous production of the indicator from contract to contract, the submission of further first article samples on the subsequent contracts may be waived at the discretion of the acquiring activity. Approval of first article samples or the waiving of first article tests does not eliminate the requirement of quality conformance inspection.

6.4.2 First article information. It shall be understood that the indicator supplied under contract shall be identical to the corresponding first article sample in material design, construction, quality, workmanship and method of manufacture. Deviation from the standards of the first article sample shall be made only by the acquiring activity. Evidence of unauthorized change shall constitute cause for rejection.

6.5 Definitions.

6.5.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

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6.5.2 Hermetic sealing. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of ceramic materials under heat or pressure and the fusion of ceramic materials, into a metallic support. Elastomeric or resinous materials, or combinations of these materials, may be used provided the leak requirements specified herein are met. Hermetic sealing does not include sealing accomplished by gaskets.

6.6 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement STANAG No. 3640, 3637, AIR STD 10/42 and 10/54. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - AS

Preparing Activity:
Navy - AS

(Project 6610-N097)

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TABLE I. Indicator operation.

CONNECTOR PINS	FUNCTION
A	Neutral (external power ground)
B	115 V 400 cps
C	Spare
D (+) E (-)	Horizontal pointer warning flag (deflects out of view)
j	Power warning flag. Connect to pin A to complete circuit
g h	Internal lighting power input 0-5 volts
d (X) e (Y) f (Z)	Roll servo control transformer stator leads
a (X) b (Y) c (Z)	Pitch servo control transformer stator leads
Y (+) Z (-)	Vertical pointer (deflects to right)
W (+) X (-)	Horizontal pointer (deflects downward)
U (+) V (-)	Vertical pointer warning flag (deflects out of view)
S (+) T (-)	Displacement pointer (deflects pointer downward)
P (+) R (-)	Rate of turn indicator (deflects to right)
L M N	
J (+) K (-)	Displacement pointer warning flag (deflects out of view)
F (X) G (Y) H (Z)	Azimuth servo control transformer stator leads

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TABLE II. Indicator operation when connected to amplifier. 1/

CONNECTOR PINS	FUNCTION
A	Azimuth follow-up test point.
B	Pitch input signal (hi) to amplifier.
C	Azimuth input signal (hi) to amplifier.
D	Pitch follow-up test point.
E	Pitch input signal (lo) to amplifier.
F	Pitch follow-up test point.
G	Roll follow-up test point.
H	18 volts A.C. excitation for generators.
J	Azimuth follow-up test point.
K	Ground.
L	Roll motor control phase.
M	Roll motor control phase.
N	115 volts, A.C., 400 cps to amplifier.
P	28 volts D.C. for power warning flag.
R	Center tap pitch trim transformer.
S	Pitch motor control phase.
T	Pitch motor control phase.
U	Center tap-motor control phase winding.
V	Roll input signal to amplifier.
W	Roll follow-up test point.
X	Azimuth motor control phase.
Y	Azimuth motor control phase.
Z	115V A.C., 400 cps motor fixed phase.
a	Spare.
b	Spare.
c	Spare.

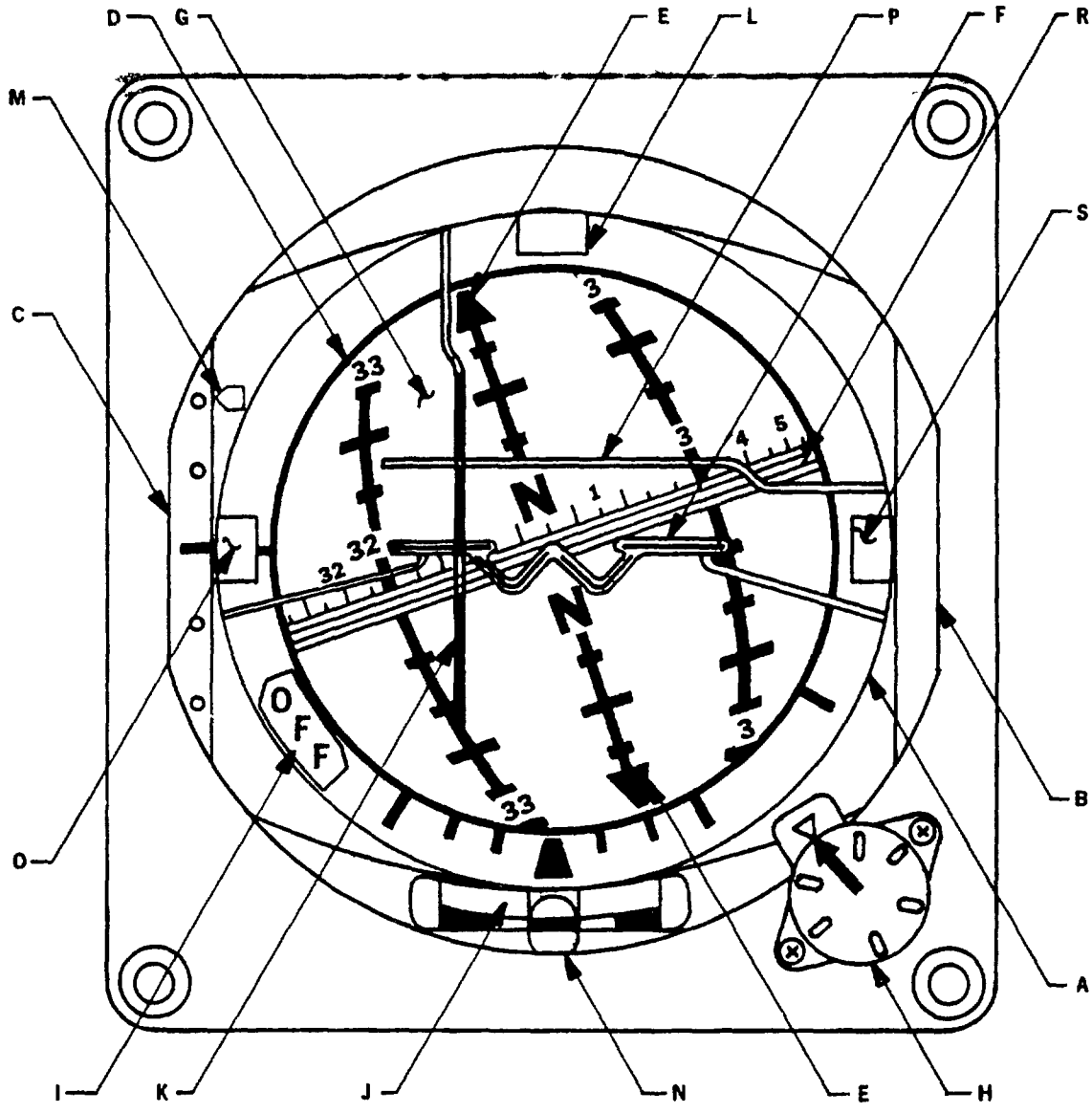
1/ The above is the identification of pins on the connector between the indicator and amplifier-power supply:

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TABLE III. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible of improper size, location, sequence or method of application; markings were not in accordance with the specification specified herein on the interior and exterior containers
Packaging and packing	Any non-conforming component; any component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

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A BANK ANGLE DIAL AND SCALE
B DIAL MASK AND VERTICAL POINTER SHIELD
C DIAL MASK AND DISPLACEMENT POINTER SCALE
D MECHANISM MASK
E BANK ANGLE INDICES
F MINIATURE AIRPLANE
G SPHERE
H PITCH TRIM KNOB
I POWER FAILURE INDICATOR
J INCLINOMETER
K VERTICAL POINTER
L VERTICAL POINTER ALARM FLAG
M DISPLACEMENT POINTER
N RATE OF TURN
O DISPLACEMENT POINTER ALARM FLAG
P HORIZONTAL POINTER
R HORIZON BAR
S HORIZONTAL POINTER ALARM FLAG

FIGURE 1.

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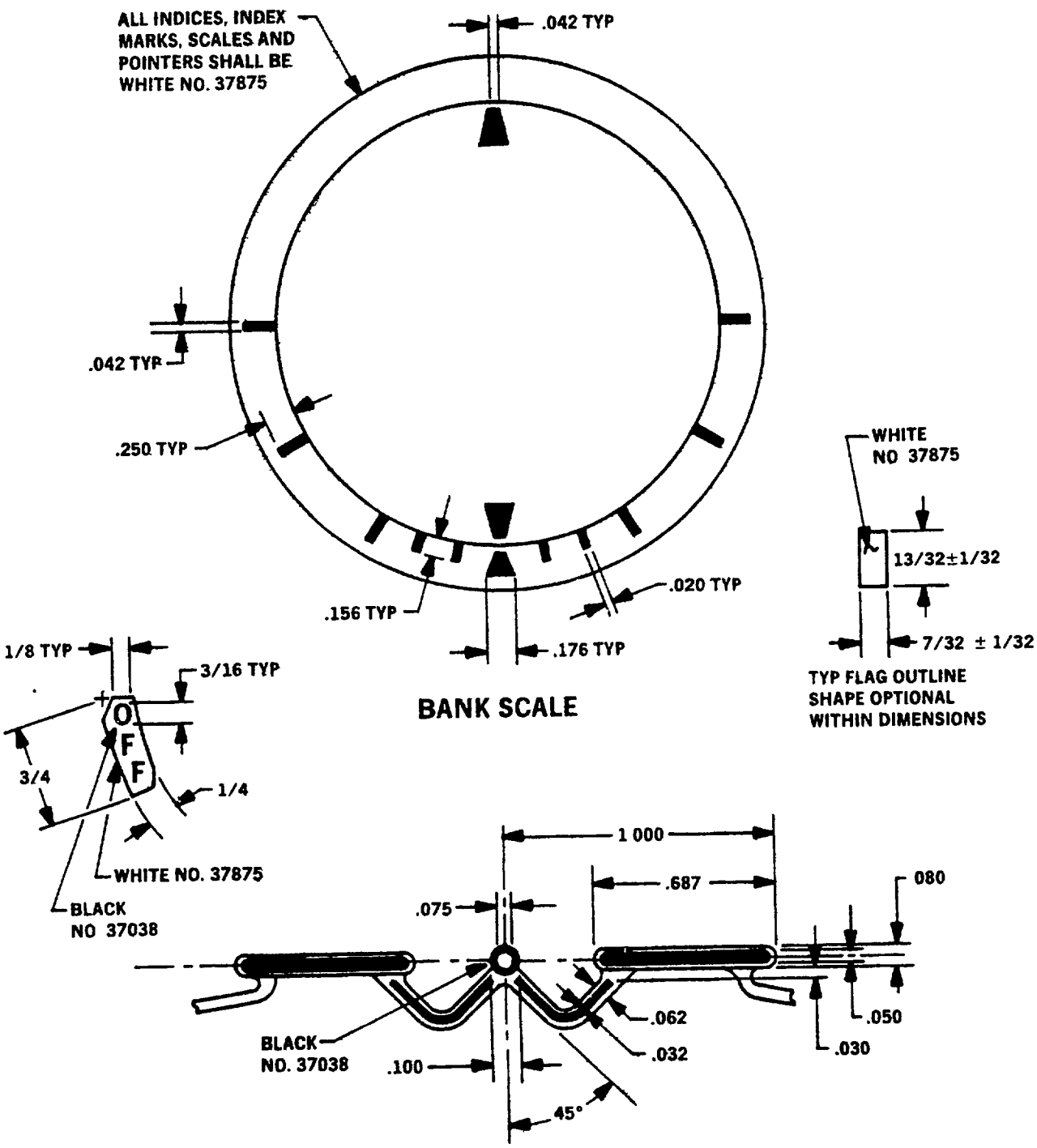
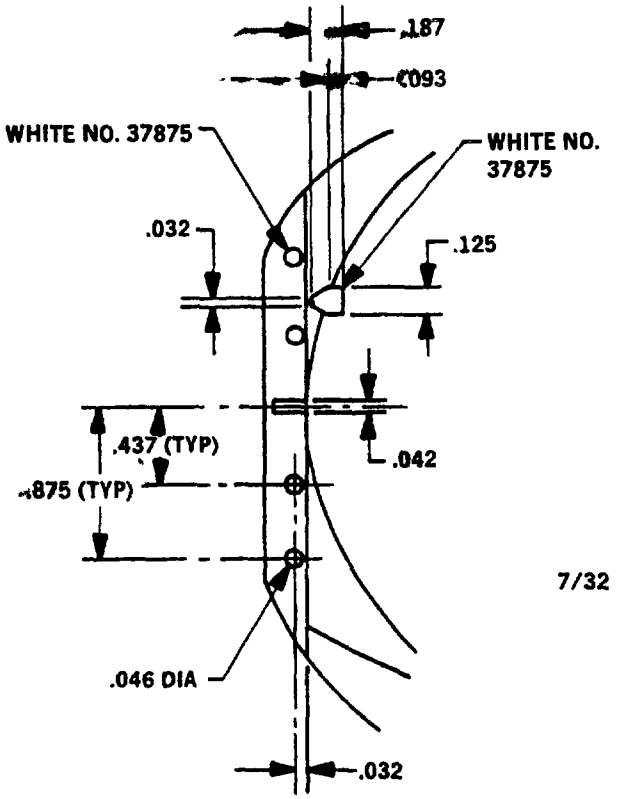
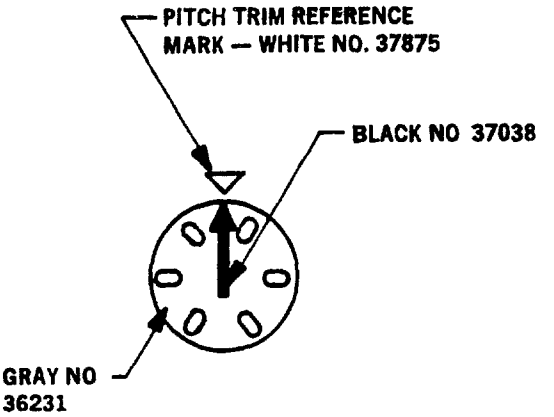


FIGURE 2.

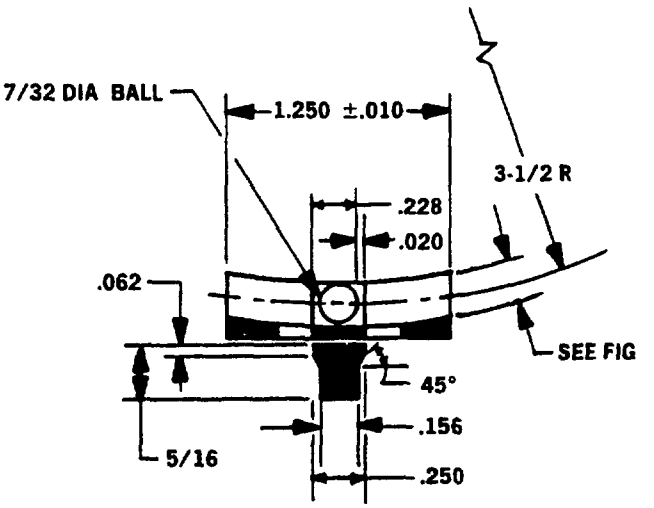
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DISPLACEMENT INDICATOR

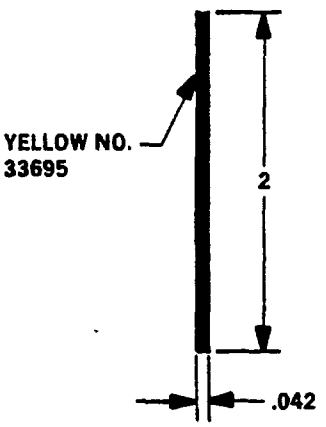


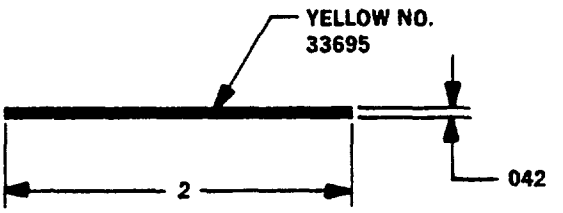
PITCH TRIM KNOB



TURN & SLIP INDICATOR

DIMENSIONS IN INCHES,
UNLESS OTHERWISE SPECIFIED
TOLERANCES·
FRACTIONS· ±1/64
DECIMALS. ±005





VERTICAL & HORIZONTAL POINTERS

FIGURE 3.

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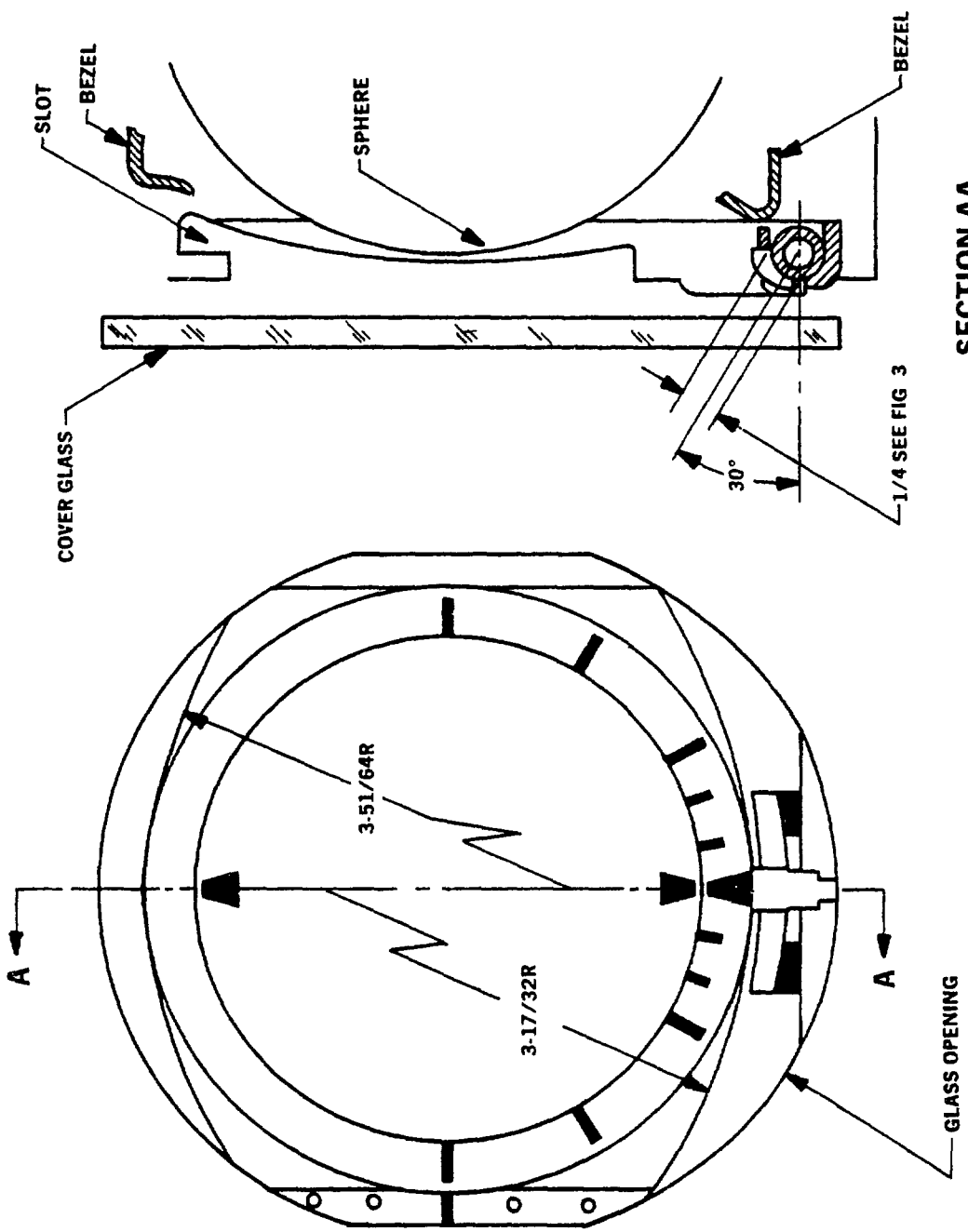
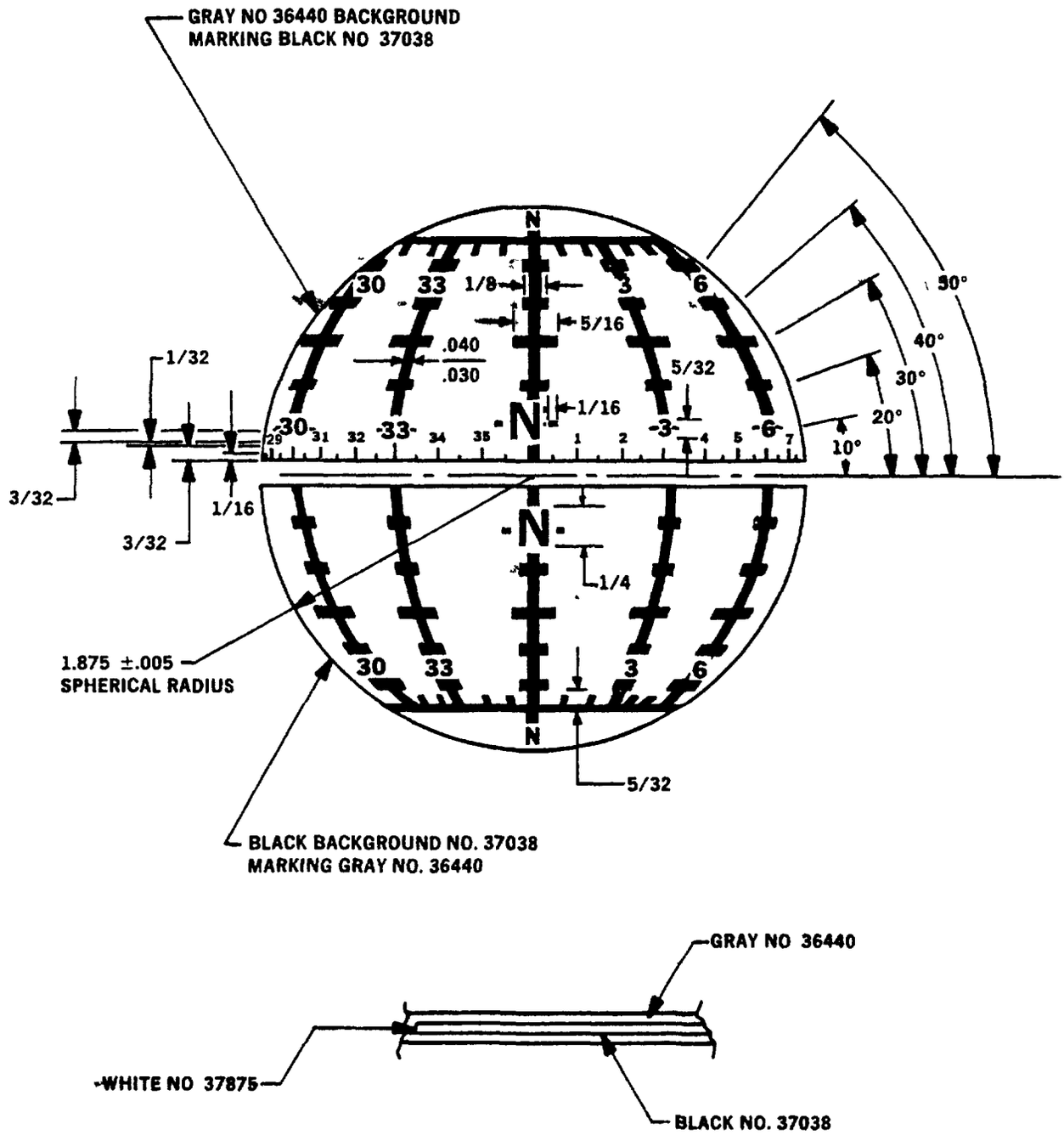


FIGURE 4.

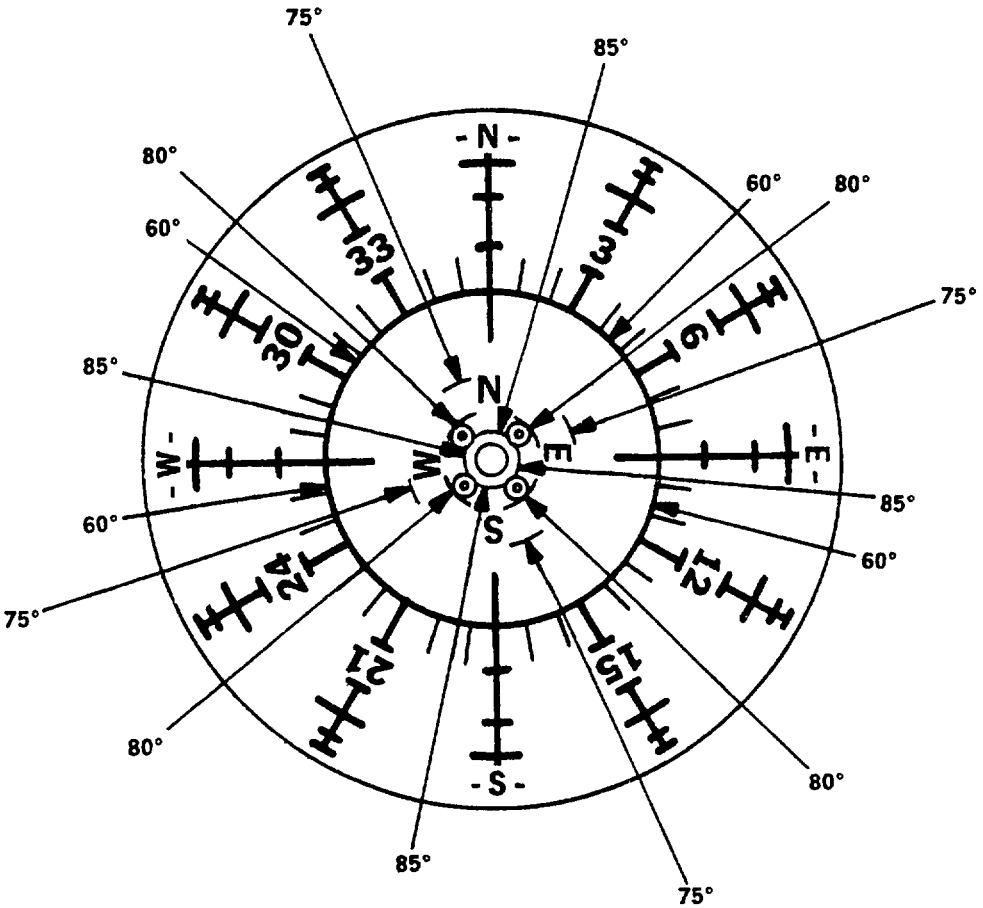
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HORIZON BAR

FIGURE 5.

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SPHERE, BOTTOM VIEW

FIGURE 6.

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Internal Wiring Diagram of Adapter

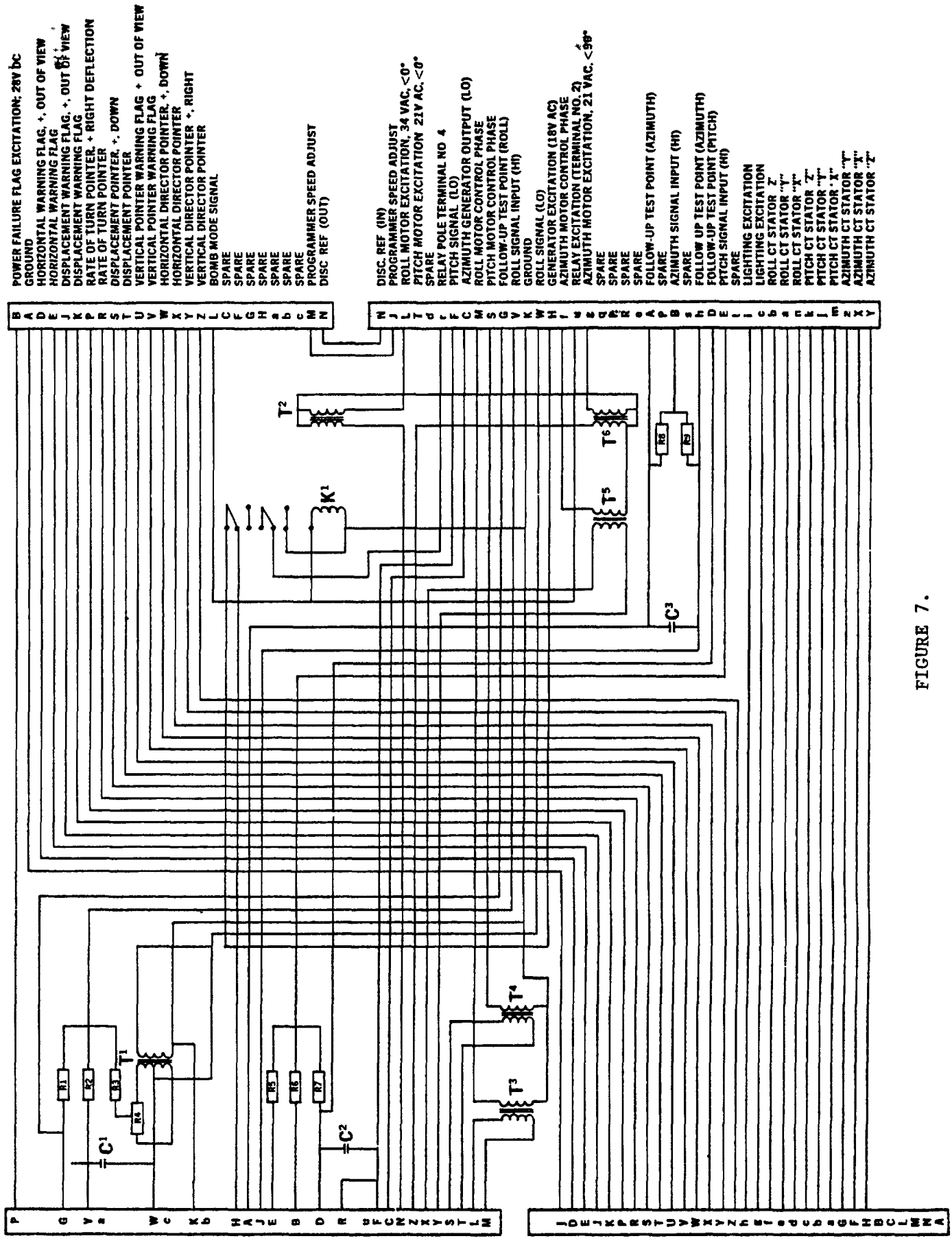


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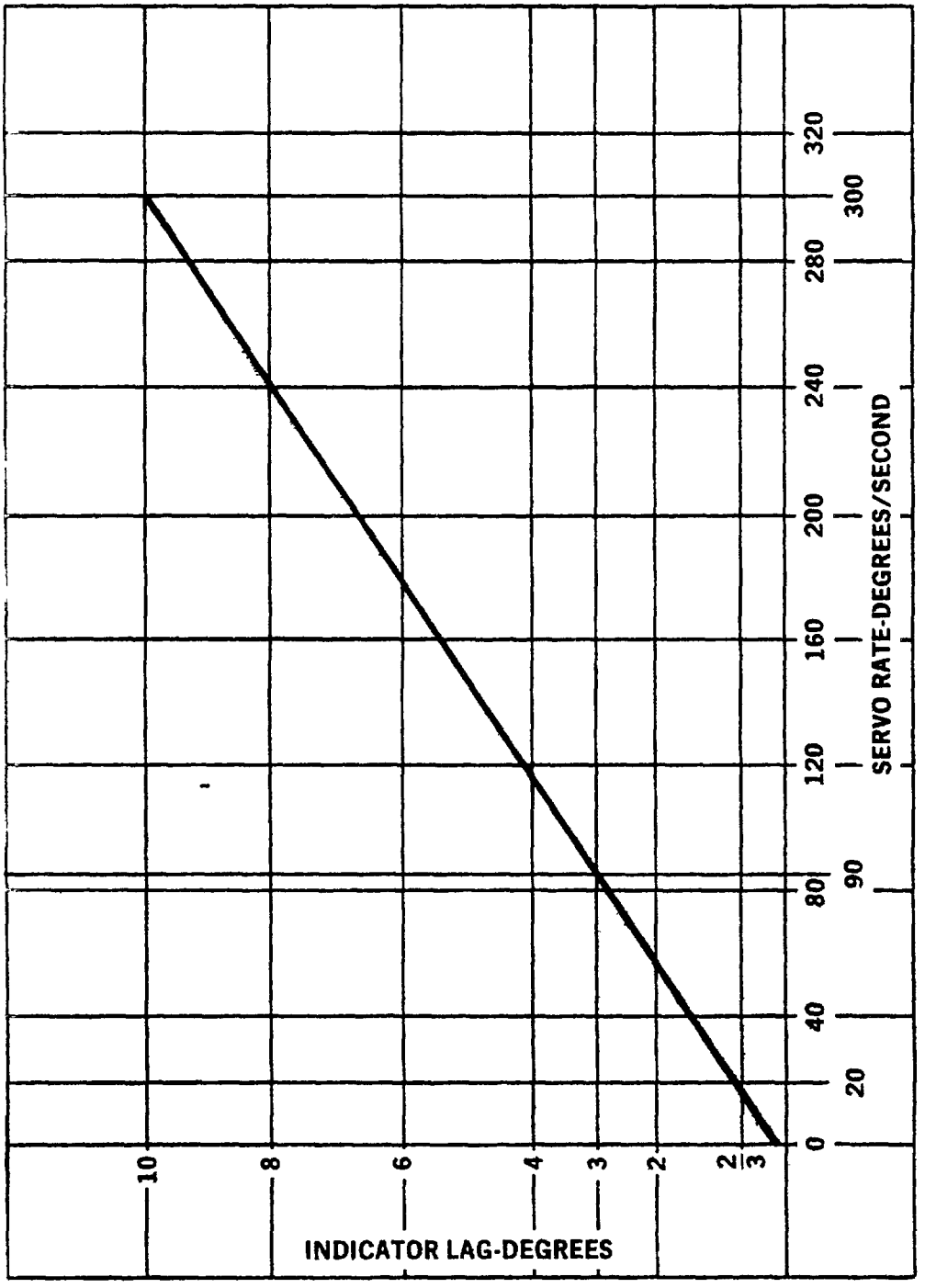


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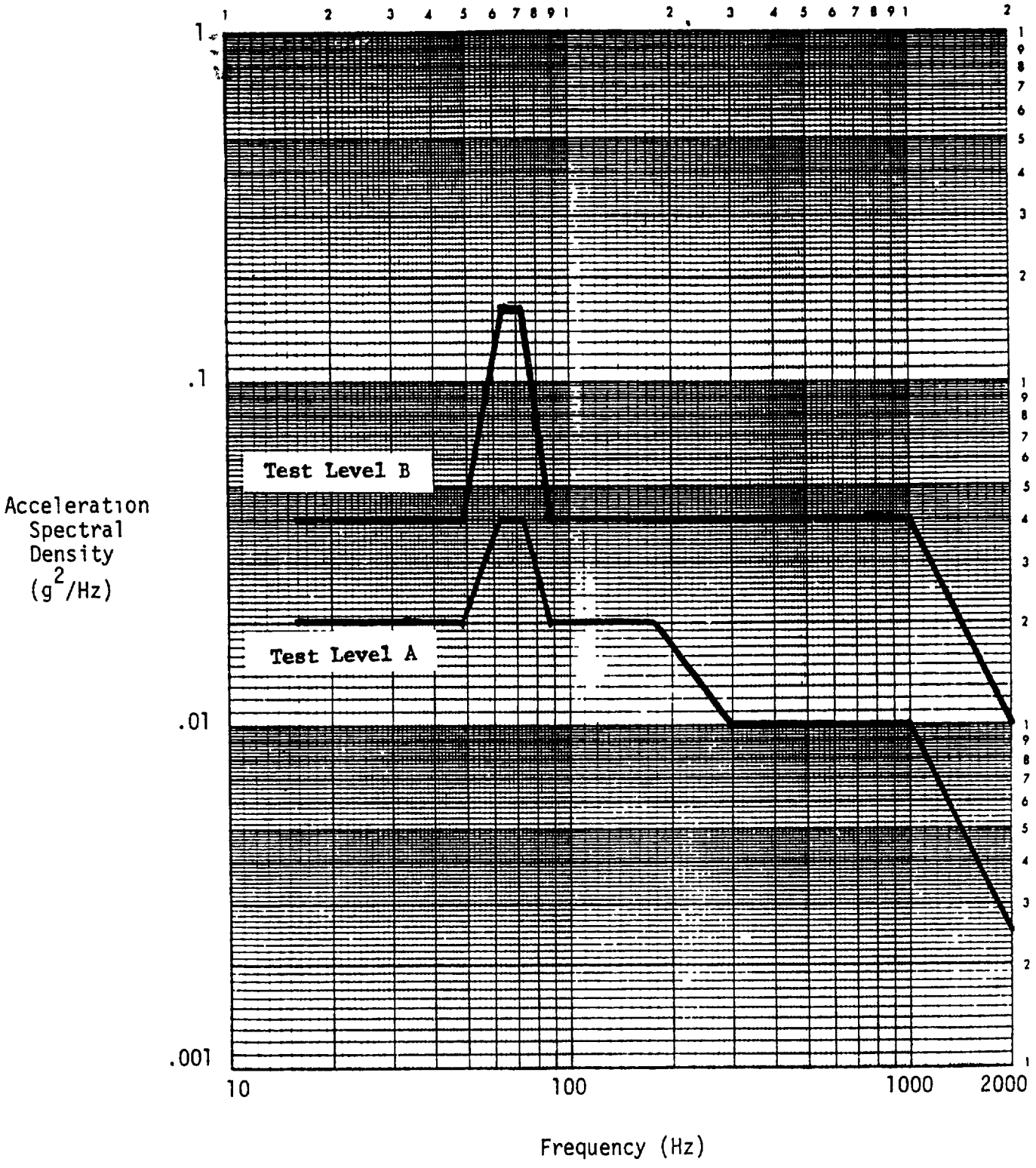


Figure 9.

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